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Chapter 28

GELFAND-P001253

17 Feb. 1986 Gelfand Seminar.

A. Beilinson [Crossed out: On the theory (?)] The Quillen determinant, strings, and so on.



Original source details: GELFAND-P001253

Uncertain reading. possibly “On the theory”

We compute the partition function

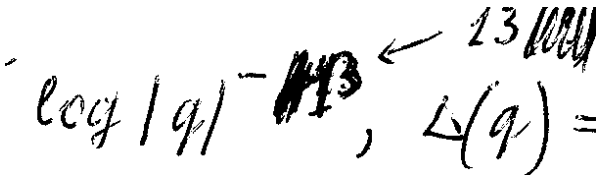
$$Z = \sum_{p \geq 0} Z_p. \tag{1}$$

p is the genus

$$Z_p = \int_{M_p} \mu_p, \qquad \begin{matrix} M_p \text{ is the moduli space of curves of genus } p. \\ \mu_p \text{ is the naturally arising measure} \end{matrix} \tag{2}$$

For $p = 1$ Kronecker (1890) and Shapiro (1972) calculated it independently (with the same result)

$$\mu_1 = \left|\frac{dq}{q}\right|^2 \Delta(q)^{-2} \log|q|^{-13}, \qquad \Delta(q) = q \prod (1-q^n)^{24}. \tag{3}$$



Original source details: GELFAND-P001253

Uncertain reading. The exponent after $\log|q|$ is heavily overwritten and reads -13. Above are 13 and illegible writing in parentheses, with an arrow to the exponent.

μ_1 is a measure on the moduli space of elliptic curves

— . —

X is a surface of genus p; g is a metric on X; L is a one-dimensional holomorphic bundle on X (holomorphic structure induced by the metric), h is a Hermitian metric on L

$$\bar\partial : \Gamma(L) \longrightarrow \Gamma(L \otimes \Omega^{0,1}). \tag{4}$$

Then

$$\underset{\text{holomorphic sections}}{H^0(X,L)} = \operatorname{Ker}\bar\partial, \qquad H^1(X,L) = \operatorname{Coker}\bar\partial = \operatorname{Ker}\bar\partial^+. \tag{5}$$

(Holomorphic bundles are described by one discrete parameter, the Chern number, and p continuous parameters, a point on the Jacobian.)

H^0, H^1 are finite-dimensional

$$d(X,L) := \frac{\det H^0(X,L)}{\det H^1(X,L)}. \tag{6}$$

$\det H^0 =$ max exterior power of H^0 is a one-dimensional vector space

[The next line has long strokes and a repetition mark, with H^1 in place of H^0.]
Hence $d(X, L)$ is also a one-dimensional vector space.
(functions on $\det H^1$ with values in $\det H^0$)
It has a natural metric

$$\left.\begin{array}{l} e_1, \dots, e_k \text{ is a basis in } H^0 \\ g_1, \dots, g_d \text{ is a basis in } H^1 \end{array}\right| \; d(X, L) \ni \lambda \frac{e_1 \wedge \dots \wedge e_k}{g_1 \wedge \dots \wedge g_d} \quad \begin{array}{l} \uparrow \\ \mathbb{C} \end{array}$$

GELFAND-P001254

[source number: 4]

The metric will be defined in two stages.
First the stupid metric. Consider the [inserted above: natural] metric on

Uncertain reading. Crossed out. Possibly “пр.”



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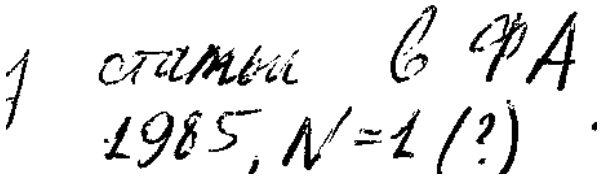
$\Gamma(L)$ and on $\Omega^{0,1}$, defining pre-Hilbert structures there.
On H^0 and $H^1 = \operatorname{Ker} \bar{\partial}^+$ metrics are also induced $\Rightarrow$
$\Rightarrow$ a metric arises in $d(X, L)$, called the stupid metric.
[Inserted above: Let us construct] the correct metric. [Downward arrow on the left.]
Consider $\Delta = \bar{\partial}^+ \bar{\partial}$ and its $\det \Delta$, calculated by Belavin and Knizhnik;
Δ' is the same Δ, with the zero modes removed.

$$\det \Delta' = \exp\left(-\zeta'_{\Delta'}(0)\right),$$

$$\zeta_{\Delta'}(s) = \sum_{\substack{\lambda \in \text{eigenvalues of } \Delta, \\ \lambda \neq 0}} \lambda^{-s}, \qquad \operatorname{Re} s > 1$$

$$\begin{aligned} &\underline{\text{Quillen metric on } d(X, L)} \quad \left(\begin{array}{c}\text{from the paper in FA} \\ 1985,\ N{=}1\ (?)\end{array}\right) \\ &\quad = (\text{stupid metric}) \cdot \sqrt{\det \Delta'}. \end{aligned} \tag{7}$$

Uncertain reading. The question mark after the issue number is in the manuscript.



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Explanation of definition (7). Let

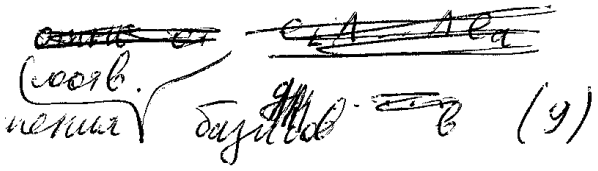
$$E_1 \xrightarrow{\bar\partial} E_2 \quad \text{be spaces with metrics (finite-dimensional)} \tag{8}$$

$$d = \frac{\det \operatorname{Ker} \bar\partial}{\det \operatorname{Coker} \bar\partial} \underset{\substack{\uparrow \\ \text{canonical} \\ \text{identification}}}{=} \frac{\det E_1}{\det E_2}. \tag{9}$$

Let bases be given. And suppose that

$$\begin{array}{l|l} e_1, \dots, e_k \text{ is a basis in } E_1 & \underbrace{e_1, \dots, e_a}_{\text{basis in } \operatorname{Ker} \bar\partial} \\ g_1, \dots, g_\ell \text{ — — " — } E_2 & g_{a+1}, \dots, g_k \\ & g_{a+1} = \bar\partial e_{a+1}, \dots, g_k = \bar\partial e_k. \end{array} \tag{10}$$

and



Uncertain reading. The crossed-out words are illegible. The product $e_1 \wedge \dots \wedge e_a$ is legible. There are more overwritten marks below.

Original source details: GELFAND-P001254

the ratios of the [inserted: corresponding] bases in (9) correspond to one another.

There are metrics on the right and on the left in (9). They do not coincide — [crossed out: difference; inserted: differ] by a factor $\sqrt{\det \Delta}$

On the left is the stupid metric, and on the right the one we need, but now we define



Uncertain reading. Crossed out. Possibly "it".

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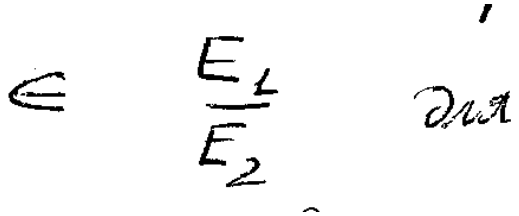
the right-hand side from the left-hand side in the infinite-dimensional situation.

GELFAND-P001255

Naturality of the "correct" metric: it does not change under variation of the operator $\bar\partial$.

In (9), if $\operatorname{Ker} \bar\partial = \operatorname{Coker} \bar\partial = 0$ on the right stands $\det \bar\partial$

$$\text{In general} \quad \det(\bar\partial : E_1 \to E_2) \in \frac{E_1}{E_2} \quad \text{for one-dimensional } E_1, E_2.$$



Uncertain reading. The quotient E_1/E_2 follows the manuscript.

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In the infinite-dimensional case there is only the modulus of the right-hand side of (9)

Unattainable ideal: define the right-hand side of (9) independently



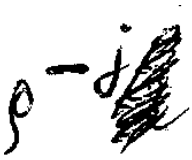
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of $\bar{\partial}$ in a functorial way (isomorphisms of spaces $\Rightarrow$ isomorphisms of determinants)

Let L be the bundle of j-differentials, $L = \omega^j$, its sections are $f(z)(dz)^j$.

$$\begin{array}{c} \text{Riemannian metric} \quad \text{[crossed-out insertion: on } L\text{]} \\ \| \\ \rho\, dz\, d\bar{z} \\ \text{induces a metric on } \omega^j \quad \Rightarrow \quad |(dz)^j|^2 = \rho^{-j}. \end{array} \tag{11}$$



Uncertain reading. Illegible signs are crossed out after $-j$.

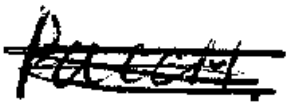
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Here $j \in \mathbb{Z}$.

The space $d(x, \omega^j)$ depends only on the complex structure on X.

The metric $|\ |_j(\underset{\substack{| \\ \text{metric on } X}}{\rho})$ is the Quillen metric on $d(X, \omega^j)$

(12) Theorem (Polyakov)



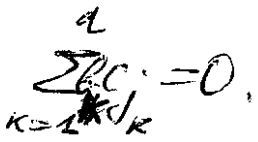
Uncertain reading. Crossed out: probably "Consider".

Original source details: GELFAND-P001255

$$\frac{\delta(\log|\ |_j(\rho))}{\delta\rho} = (6j^2 - 6j + 1) \qquad \underset{\substack{| \\ \text{curvature of the metric}}}{R(\rho)}$$

(13) Notation: $c_j = 6j^2 - 6j + 1$ (second Bernoulli polynomial)

(14) Corollary. Let there be a collection $j_1, \ldots, j_a \in \mathbb{Z}$, [inserted above: and integers $b_1, \ldots, b_a \in \mathbb{Z}$,] such that $\sum_{k=1}^{a} b_k c_{j_k} = 0$.



Uncertain reading. The subscript on b is overwritten. The summation and the exponent in the next line support reading it as k.

Original source details: GELFAND-P001255

Consider the Quillen metric on $\otimes d(X, \omega^{j_k})^{b_k}$.



Uncertain reading. A short crossed-out passage is illegible.

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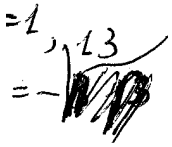
It is then independent of the [inserted: original] ρ, and depends only on the complex structure on X

(15) Definition. c_j is called the conformal anomaly.

(16) Most important example.

$$j_1 = 2, \quad b_1 = 1,$$
$$j_2 = 1, \quad b_2 = -13$$

— the corresponding number 13



Uncertain reading. In b_2, the earlier writing after the minus sign is overwritten. The insertion above reads 13.

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(In 26-dimensional space there is no conformal anomaly)

— . —

GELFAND-P001256

[source number: 5]

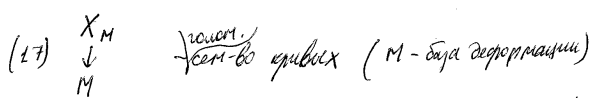


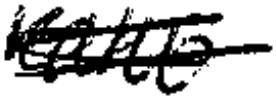
Diagram and typed description.

(17) $\begin{array}{c} X_M \\ \downarrow \\ M \end{array}$ [inserted: holomorphic] family of curves (M is the base of deformations)

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Then $\forall$ integer j there arises a one-dimensional bundle d_j over M with fiber $d(X_t, \omega^j)$ over $t \in M$.

This is



Uncertain reading. Crossed out: probably “is”.

Original source details: GELFAND-P001256

a one-dimensional holomorphic bundle over M.

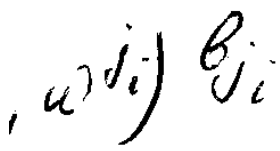
(18)



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over $\otimes d(X_t, \omega^{j_i})^{b_{j_i}}$ there is a canonical Hermitian metric (b_j as in (14))



Uncertain reading. Here the exponent is written b_{j_i}.

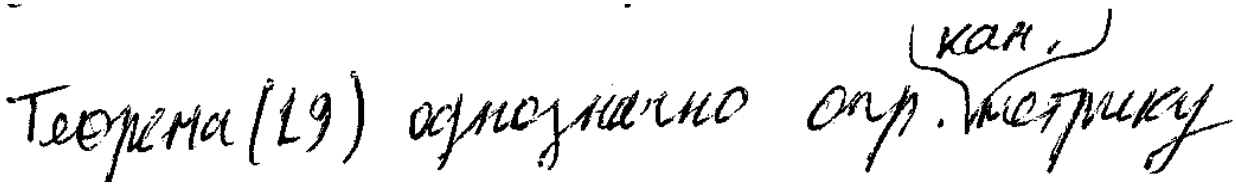
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(19) Theorem (Belavin, Knizhnik):

The curvature of the canonical metric on $\otimes d(X, \omega^{j_i})^{b_i}$ equals 0.

In words: Cancellation of the conformal anomaly implies cancellation of the gravitational one.

Theorem (19) uniquely determines the [inserted: canonical] metric (which will be determined by the genus alone)



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Uncertain reading. The statement of uniqueness follows the manuscript.

Belavin

$$\sum_{\text{surfaces}} e^{-a(\text{area})} = \sum_{p=2}^{\infty} \int d\mu_0^{(p)} \frac{\det \Delta_{-1}^{(p)}}{(\det \Delta_0^{(p)})^{13}} \qquad (20)$$

(Polyakov + refinements for arbitrary genus)

Δ_j is the Laplacian on j-forms

Important: on the right is

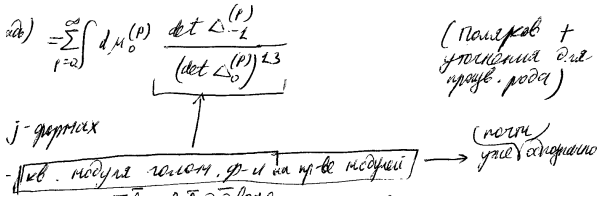


Diagram and typed description.

$$\underbrace{\frac{\det \Delta_{-1}^{(p)}}{(\det \Delta_0^{(p)})^{13}}}$$

↑

squared modulus of a holomorphic function on the moduli space

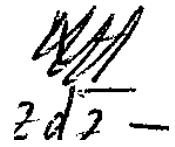
↓

already [inserted: almost] unique

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$$\delta\bar{\delta} \log \det \Delta_j = c_j \int \frac{\partial f\, \bar{\partial} \bar{f} - f \bar{f}\, \partial\bar{\partial} \log \rho}{\rho^2}, \qquad f = \rho\eta, \qquad (21)$$

$$\delta_\rho \log \det \Delta_j = c_j R \qquad \text{Variation of the metric: } dz\, d\bar{z} \longrightarrow |dz + \eta\, d\bar{z}|^2 \qquad (21')$$



Uncertain reading. Illegible signs are crossed out above $d\bar{z}$ to the left of the arrow.

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Beilinson

The integral in (20) diverges and still has to be renormalized.

Belavin

$$Z_1 = \int \frac{d\tau\, d\bar{\tau}}{(\operatorname{Im}\tau)^2} \cdot \frac{1}{(\operatorname{Im}\tau)^{12}} \cdot \frac{1}{|\Delta(q)|^2}, \qquad q = e^{2\pi i \tau}$$



Uncertain reading. An illegible sign is crossed out after $(\operatorname{Im}\tau)^2$ in the first denominator.

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The integral diverges.

But there is a conjecture that everything converges in the superstring.

$$Z_2 = \int \frac{d^2\tau}{(\operatorname{Im}\tau)^3} \cdot \frac{1}{(\operatorname{Im}\tau)^{10}} \cdot \frac{1}{|F(\tau)|^2}, \qquad \tau = \begin{bmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{bmatrix}$$

τ is the period matrix

(Knizhnik, Morozov, Perelomov)

$$F(\tau) = \prod \theta_m^2(\tau)$$

Z_3 exists as well.

For general p there is no good description of the moduli space.

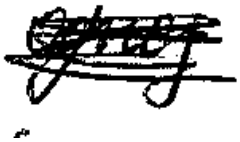
GELFAND-P001257

Beilinson

M is the moduli space.

$$p > 2 \Rightarrow \forall \text{ holomorphic function} = \text{const}$$

A metric [inserted above: of zero curvature] is determined up to the squared modulus of



Uncertain reading. Crossed out: possibly "function".

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a holomorphic function $\Rightarrow$ thus uniquely. It remains to calculate it. (This reduces to calculating the unique holomorphic section



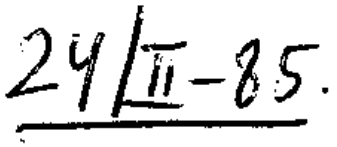
Uncertain reading. There is a short overwritten mark beneath the abbreviation for "holomorphic". Its reading is uncertain.

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of a certain bundle).

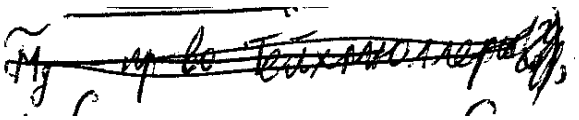
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24 Feb. 85. Kontsevich (Maxim) (Gelfand Seminar)



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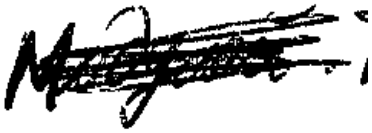
Uncertain reading. The manuscript reads "24/II-85". Catalogue record 667 dates the talk 24 February 1987.



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Uncertain reading. Crossed out: $\widetilde{M}_g$ is the space infinite-dimensional... [ending overwritten].

M_g is the space of



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Uncertain reading. Crossed out: probably "moduli".

Teichmüller

$$(\text{Curve } C,\ \text{point } p \in C,\ x \text{ is a coordinate at } p) = \widetilde{M}_g$$

$$\mathfrak{g} = \mathbb{C}((x))\frac{\partial}{\partial x}$$

$$\cup$$

$$\mathfrak{k} = \mathbb{C}[[x]]\frac{\partial}{\partial x}$$

γ is the Laurent series of vector fields on C with a pole at p.

$\gamma \subset \mathfrak{g}$



Original source details: GELFAND-P001258

Uncertain reading. The first γ is retraced and overwritten. It is repeated clearly below.



Original source details: GELFAND-P001258

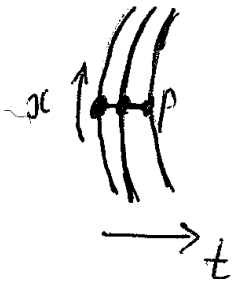
Uncertain reading. The ending of the word translated as "with a pole" is retraced and overwritten. A similar expression below supports the reading.

$\widetilde{M}_g$ in a neighborhood of a point is the homogeneous space $\mathfrak{g}/\gamma$

(1) Thm. 1. The algebra $\mathfrak{g}$ acts on $\widetilde{M}_g$, the stabilizer of the point [inserted above: t] is γ

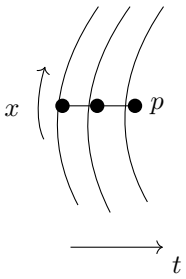
Description of the action of $\mathfrak{g}$ on $\widetilde{M}_g$

Let a one-parameter family of curves be given. On this nonlinear bundle there is a connection with a pole only at the point p.



Original source details: GELFAND-P001258

Diagram and typed description.



$$\nabla_t = \frac{\partial}{\partial t} + F(x,t)\frac{\partial}{\partial x}$$

The tangent space to $\widetilde{M}_g$ is identified with $\mathfrak{g}/\mathfrak{k}$.



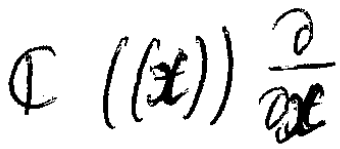
Original source details: GELFAND-P001258

Uncertain reading. The denominator in this line is $\mathfrak{k}$.

$$T(\widetilde{M}_g)_{(C,x,p)} \underset{\substack{\uparrow \\ \text{isomorphism} \\ \text{of Kodaira–Spencer}}}{=} \mathbb{C}((x))\frac{\partial}{\partial x}/$$

vector fields extending to $C \setminus \{p\}$

[In the margin: (all of C!!)]



Original source details: GELFAND-P001258

Uncertain reading. The x in the Laurent series and in the derivative denominator is retraced. The earlier strokes are illegible.

$\widetilde{\mathfrak{g}}$ is the Virasoro algebra. It acts



Original source details: GELFAND-P001258

Uncertain reading. Crossed out: in $\widetilde{M}_g$ [part of the subscript is overwritten].

in the bundle d_k.

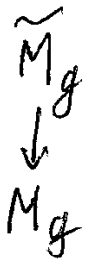


Diagram and typed description.

$$\begin{array}{c}\widetilde{M}_g\\ \downarrow\\ M_g\end{array}$$

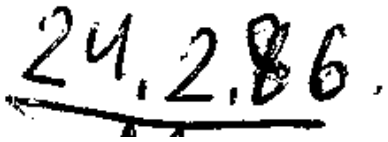
Original source details: GELFAND-P001258

d_k is the determinant bundle over M_g, lifted to $\widetilde{M}_g$

GELFAND-P001259

24 Feb. 86. Gelfand Seminar

Morozov Consequences of the Belavin–Knizhnik theorem.



Uncertain reading. The manuscript reads “24.2.86”. Catalogue record 668 dates the talk 24 February 1987.

Original source details: GELFAND-P001259

Explicit formulas for bosonic strings, etc.

The bosonic string in the critical dimension $d = 26$.

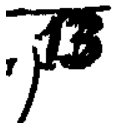
Summary of Beilinson’s talk:



Uncertain reading. The first word overlaps strokes from the line above. The reading is “Summary”.

Original source details: GELFAND-P001259

$$\sum_p \int Dx\, Dg\ \exp \int \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu\, d^2\xi \prod_\alpha V_\alpha = \\ = \sum_p \int_{M_p} \frac{|d\mu_p|^2}{(\det \operatorname{Im} T)^{13}}\, K\{V\} \tag{1}$$



Uncertain reading. The exponent 13 is heavily retraced.

Original source details: GELFAND-P001259

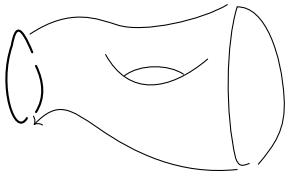
V_α are vertex operators (this will be explained later)



Original source details: GELFAND-P001259

Diagram and typed description.

(2)



worldsheet of a propagating little ring

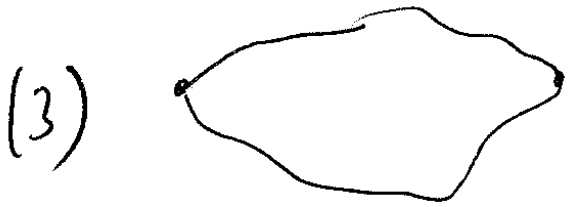


Original source details: GELFAND-P001259

Uncertain reading. The abbreviation is read as “распр.”, expanded in the translation as “propagating”.

Mutual transformations are accounted for by considering different topologies.

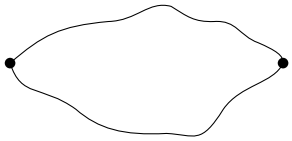
Technically it is convenient to take points at the beginning and end.



Original source details: GELFAND-P001259

Diagram and typed description.

(3)



Then V_α are δ-functions. Taking the Fourier transform, we obtain (α is the number of the vertex operator)

$$V_\alpha(p) = \int e^{ip_\alpha x(\xi)} d^2\xi \tag{4}$$



Original source details: GELFAND-P001259

Uncertain reading. An illegible sign, possibly μ, is crossed out after p_α in the exponent.

It is also convenient to take derivatives of δ-functions:

$$V_\alpha(p) = \int \partial x^\mu(\xi) e^{ip_\alpha x(\xi)} d^2\xi \tag{4'}$$

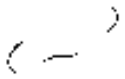
To obtain the right-hand side of (1):

a) Integrate over x to obtain

$$\frac{1}{(\det \Delta_0)^{13}} \prod_\alpha \int d^2\xi_\alpha \prod_{\alpha,\beta} \exp p_\alpha p_\beta \, G_{(reg)}(\xi_\alpha, \xi_\beta) \tag{5}$$

Δ_0 is the Laplacian on scalars

$G_{(reg)}$ is the regularized Green's function (smoothed when the points coincide)



Uncertain reading. Faint curved strokes appear in the right margin. Their reading is uncertain.

Original source details: GELFAND-P001259

GELFAND-P001260

[source number: 11]

b) Integrate over metrics.

$$g^{ab} = e^\varphi \delta^{ab} + (\text{traceless contributions}) \tag{6}$$



Uncertain reading. The ending of the first word in "traceless contributions" is overwritten.

Original source details: GELFAND-P001260

Eliminate the latter by the transformations

$$\xi \longrightarrow \xi + \varepsilon$$
$$g \longrightarrow g + \partial G$$

There arises $\det \Delta_{-1}$. We arrive at

$$\int d\mu_p \frac{\det \Delta_{-1}}{(\det \Delta_0)^{13}} \tag{7}$$

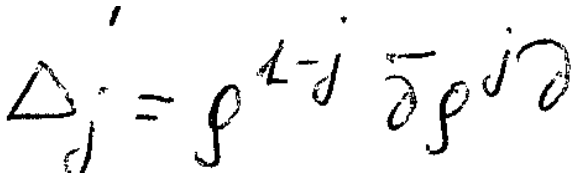
In the critical dimension, dependence on φ disappears.
Accounting for zero modes.

$$N^{(j)}_{\alpha\beta} = \langle \omega^{(j)}_\alpha \omega^{(j)}_\beta \rangle = \int \rho^{1-j} \bar{\omega}_\alpha \omega_\beta \, d^2\xi \tag{8}$$

There are j-differentials: $f(\xi) d\xi^j$,
the corresponding Laplace operator

$$\Delta_j = \rho^{1-j} \bar{\partial} \rho^j \partial \tag{9}$$

The zero modes are holomorphic differentials.

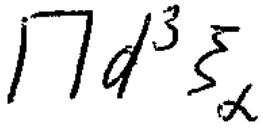


Original source details: GELFAND-P001260

Uncertain reading. In (9) the first ∂ bears a bar and the second is written without one.

Hence $\det N^{(j)}$ is defined.
We obtain

$$\int |d\nu_p|^2 \left(\frac{\det N^{(0)}}{\det' \Delta_0} \right)^{13} \det{}' \Delta_{-1} \prod d^3\xi_\alpha \prod_{\alpha,\beta} \exp p_\alpha p_\beta \, G_{(Reg)}(\xi_\alpha, \xi_\beta) \tag{10}$$

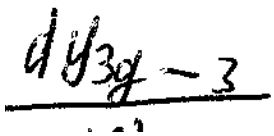


Original source details: GELFAND-P001260

Uncertain reading. Equation (10) reads $d^3\xi_\alpha$.

Here

$$d\nu_p = \frac{dy_1 \ldots dy_{3g-3}}{\det N^{(2)}} \tag{11}$$



Original source details: GELFAND-P001260

Uncertain reading. The last subscript in the numerator reads $3g-3$.

in the proper holomorphic coordinates on the moduli space.

$$dz\,d\bar z + f(z)dz^2 + f(\bar z)d\bar z^2 \qquad (12)$$
$$\| \quad \text{On the given surface } g\,dz\,d\bar z;$$



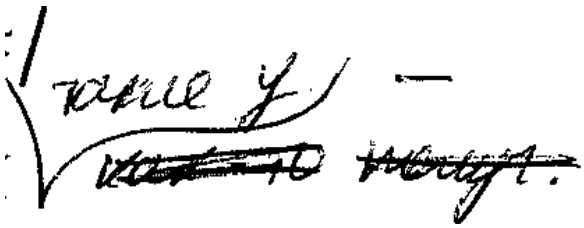
Uncertain reading. The coefficient of $dz\,d\bar z$ is retraced. It reads g.

Original source details: GELFAND-P001260

on neighboring surfaces

$$\rho\,|dz + y\eta\,d\bar z|^2 \qquad (13)$$

(14) Ahlfors–Bers theorem:

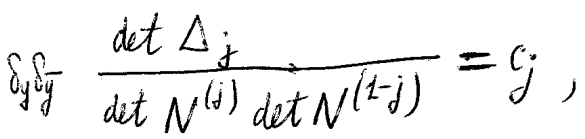


Uncertain reading. Crossed out: “some coord.” (the first word is partly illegible).

Original source details: GELFAND-P001260

[Inserted above: such y are] holomorphic coordinates on the moduli space
y parametrizes the deviation from the surface.
(15) Assertion. (Belavin–Knizhnik theorem)

$$\delta_y\delta_{\bar y}\frac{\det\Delta_j}{\det N^{(j)}\det N^{(1-j)}} = c_j, \qquad c_j = 6j^2 - 6j + 1. \qquad (16)$$



Uncertain reading. In (16) the operators $\delta_y\delta_{\bar y}$ stand directly before the quotient. No log is written.

Original source details: GELFAND-P001260



Uncertain reading. Several illegible mathematical signs are crossed out after $c_j =$.

Original source details: GELFAND-P001260

GELFAND-P001261

Gelfand’s question: there is no proper theory of det for elliptic operators



Uncertain reading. The abbreviation before “theory” is “прав.”, translated here as “proper”.

Original source details: GELFAND-P001261

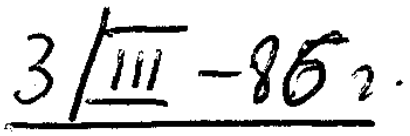
$$\det(AB) \stackrel{?}{=} \det A \cdot \det B$$

(A, B are elliptic operators of arbitrary order, invertible)

GELFAND-P001262

3 Mar. 86. Gelfand Seminar

Beilinson Mixed Hodge structures



Uncertain reading. The manuscript reads "3/III—86". Catalogue record 669 dates the talk 3 March 1987.

Original source details: GELFAND-P001262

(1) Def. A mixed Hodge structure is

1. A finitely generated abelian group L

and two filtrations

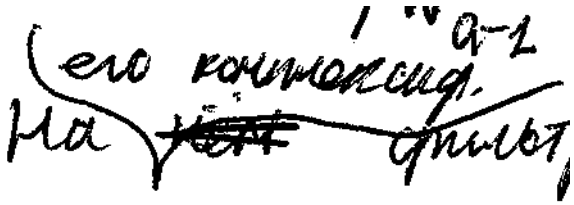
2. $L_{\mathbb{Q}}$: $\quad \ldots \subset W_i \subset W_{i+1} \subset \ldots$

3. $L_{\mathbb{C}}$: $\quad \ldots \supset F^j \supset F^{j+1} \supset \ldots$

Conditions:

W_a/W_{a-1} is a vector space over $\mathbb{Q}$.

On



Uncertain reading. Crossed out: probably "it".

Original source details: GELFAND-P001262

[inserted above: its complexification] filtration



Uncertain reading. Crossed out: possibly (3).

Original source details: GELFAND-P001262

$(1)_3$ induces a filtration:

$$F^j_{(a)} = \frac{F^j \cap (W_a)_{\mathbb{C}}}{(W_{a-1})_{\mathbb{C}} \cap F^j} \qquad \text{is the filtration on } (W_a/W_{a-1})_{\mathbb{C}}. \tag{4}$$

Required: $\forall a, j$

$$F^j_{(a)} \oplus \bar{F}^{a+1-j}_{(a)} \xrightarrow{\sim} (W_a/W_{a-1})_{\mathbb{C}} \tag{5}$$

or equivalently

$$\underline{\text{subspaces}}\ (W_a/W_{a-1})^{p,a-p}_{\mathbb{C}} = F^p_{(a)} \cap \bar{F}^{a-p}_{(a)} \tag{6}$$



Uncertain reading. The intersection sign is heavily retraced.

Original source details: GELFAND-P001263

as one varies p they define a grading on $(W_a/W_{a-1})_{\mathbb{C}}$

Another definition:

instead of $(1)_2$ and $(1)_3$ take the pair

$(7)_2$ A bigrading on $L_{\mathbb{C}}$: $L_{\mathbb{C}} = \bigoplus L^{p,q}$



Uncertain reading. The sign is overwritten. It reads $=$.

Original source details: GELFAND-P001263

$(7)_3$ $N \in \text{End}\, L_{\mathbb{C}}$

and the following properties must hold

(8) a) $\bar{L}^{p,q} = L^{q,p}$

b) $\bar{N} = -N$

c) $N(L^{p,q}) \subset \bigoplus_{\substack{a<p \\ b<q}} L^{a,b}$

d) the filtration



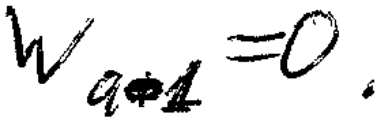
Undeciphered passage. Illegible crossed-out writing.

Original source details: GELFAND-P001263

$$W_i := \bigoplus_{p+q\le i} L^{p,q}$$

is rational (it is real by condition a)), that is, defined over $\mathbb{Q}$.

(9) Example 1. C is a smooth complex manifold. On $H^a(C,\mathbb{Z})$ there is a pure Hodge structure of weight a, that is, $W_a = H^a(C,\mathbb{Z})$, $W_{a+1} = 0$.

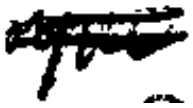


Uncertain reading. The sign before 1 is overwritten and resembles a plus. The next line clearly has $W_{a-1} = 0$.

Original source details: GELFAND-P001263

(10) (def. L is a pure Hodge structure of weight a, if $W_{a-1} = 0;\ W_a = L_{\mathbb{Q}}$).

In

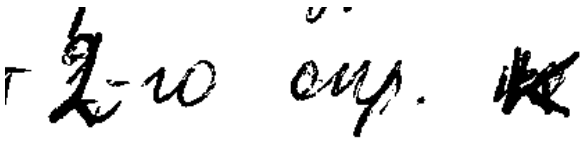


Undeciphered passage. Illegible crossed-out writing.

Original source details: GELFAND-P001263

Example 1 we have $N = 0$.

Passage from the second definition to the first:



Uncertain reading. The numeral 2 is retraced. The preposition “to” is overwritten.

Original source details: GELFAND-P001263

$$F^i = \exp(N)\Big(\bigoplus_{\substack{p\ge i \\ q \text{ arbitrary}}} L^{p,q}\Big) \tag{11}$$

$(10')$ def. 10 in the second version: L is a pure structure of weight a, if $L = \bigoplus_{p+q=a} L^{p,q}$

(12) In this case $N = 0$ (this follows from condition (8) c)).

GELFAND-P001264

[source number: 14]

(13) Example 2. C is a compact Riemann surface with four given points x_1, x_2, y_1, y_2.

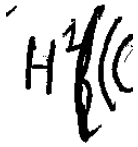


Uncertain reading. The ending of the word for “points” is retraced.

Original source details: GELFAND-P001264

On

$$H^1_{\mathbb{Q}}(C \setminus \{x_1, x_2\}, \{y_1, y_2\}) \tag{14}$$



Uncertain reading. The retraced subscript on H^1 reads $\mathbb{Q}$.

Original source details: GELFAND-P001264

there is a mixed Hodge structure:

$$W_0 \subset W_1 \subset W_2 \tag{15}$$

$$W_0 = \mathbb{Z}_{\mathbb{Q}}, \quad W_1/W_0 = H^1(C)_{\mathbb{Q}}, \quad W_2/W_1 = \mathbb{Z}_{\mathbb{Q}}, \tag{16}$$

[At right:]

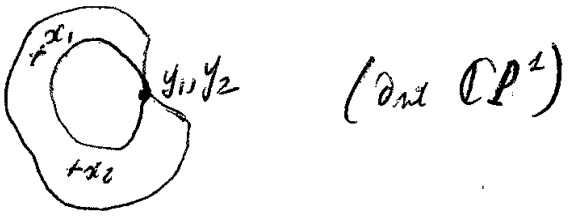
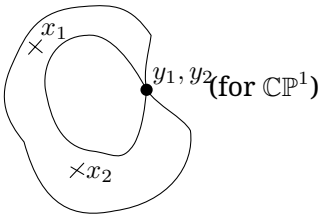


Diagram and typed description.



Original source details: GELFAND-P001264

namely

$$W_0 = \mathrm{Ker}\Big(\underset{\substack{| \\ \text{the group } (14)}}{H^1} \longrightarrow H^1(C \setminus \{x_1, x_2\}) \Big) \tag{17}$$

In terms of forms:

$$H^1 = \frac{\text{closed 1-forms on } C \setminus \{x_1, x_2\}}{\{d(f) : f(y_1) = f(y_2)\}} \tag{18}$$



Uncertain reading. The letter C in the numerator of (18) is retraced.

Original source details: GELFAND-P001264

$$W_1 = \mathrm{Ker}(H^1 \xrightarrow{\mathrm{Res}_{x_1}} \mathbb{Z}) \tag{19}$$

$$W_2 = H^1 \tag{20}$$

Passage to the second definition: On W_0 there is only type $(0,0)$

On $W_1/W_0 = H^1(C)$ there are types $(1,0)$ and $(0,1)$

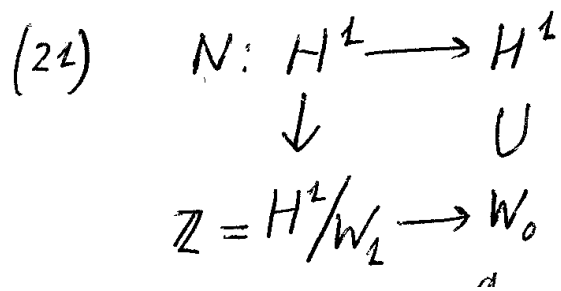


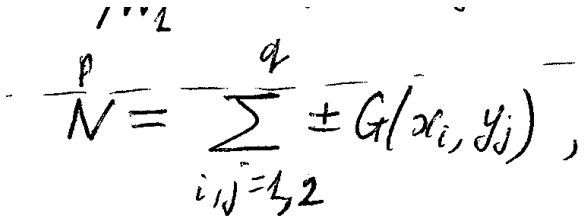
Diagram and typed description.

$$(21)\quad N: \begin{array}{ccc} H^1 & \longrightarrow & H^1 \\ \downarrow & & \cup \\ \mathbb{Z} = H^1/W_1 & \longrightarrow & W_0 \end{array}$$

Original source details: GELFAND-P001264

— here there is also a natural basis

$$\overset{p}{N} = \sum_{i,j=1,2}^{q} \pm G(x_i, y_j), \tag{22}$$



Uncertain reading. The letters p and q stand above (22), close to the broken horizontal line.

Original source details: GELFAND-P001264

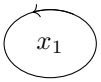
where G is the Green's function of the surface C (the Green's function of the Laplacian on functions) for some metric

Standard bases:

$$p: \quad \int p = 1 \qquad \text{is a basis in } H^1/W_1$$



Diagram and typed description.



Original source details: GELFAND-P001264

$$q: \quad \int_{y_1}^{y_2} q = 1 \qquad \text{is a basis in } W_0$$

N is written in (22) in the bases p, q.

GELFAND-P001265

Varchenko applied this to calculate the asymptotics of oscillatory integrals (see [inserted above: his] paper in Izv. Akad. Nauk, about four years ago)

Morozov. Strings (continued)



Undeciphered passage. An illegible word is crossed out below the heading.

Original source details: GELFAND-P001265

Size of the hypothetical string $\sim 10^{-33}$ cm

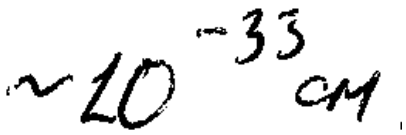
(Cf. the size of a proton



Undeciphered passage. The crossed-out passage is illegible.

Original source details: GELFAND-P001265

is the same $\sim 10^{-33}$ cm.



Original source details: GELFAND-P001265

Uncertain reading. Comparison with the two other occurrences on the leaf supports reading the exponent as -33.

—"— of an electron $\sim 10^{-18}$ cm)

All existing theories, except string theory, are internally inconsistent because of ultraviolet divergences

Measurable quantities should be independent of renormalizations. But everywhere at distances of $\sim 10^{-33}$ cm divergences occur (the inverse quantity is the Planck mass M_P)

$$V(k) = \int d^2\xi \, e^{ikx(\xi)} \tag{1}$$

$$\langle V(k_1) \dots V(k_n) \rangle = \int Dx \, Dg \, e^{-\sqrt{g}\, g^{ab} \partial x \partial x} V(k_1) \dots V(k_n) \tag{2}$$

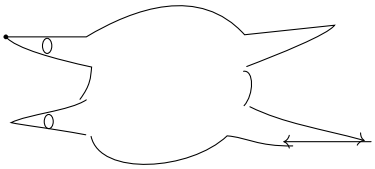
interpret as a scattering amplitude



Original source details: GELFAND-P001265

Diagram and typed description.

(3)



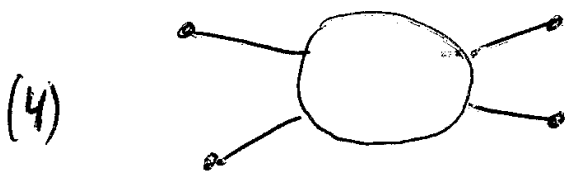
— physical picture

In the critical dimension it can be contracted



Original source details: GELFAND-P001265

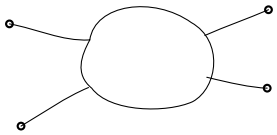
Uncertain reading. The last word is read as "contracted".



Original source details: GELFAND-P001265

Diagram and typed description.

(4)



It behaves like a small spring tending to contract to a point.

The main delight: the correlators will be gauge invariant.

$$V_\mu(k) = \int d^2\xi\, \partial x(\mu) e^{ikx(\xi)} \tag{5}$$

$$\langle V_{\mu_1}(k_1) \dots V_{\mu_n}(k_n) \rangle \tag{6}$$

have the property of gauge invariance!! and describe the interaction in Yang–Mills theory (7) + consistent with gravity!



Uncertain reading. The word after the plus sign reads “согласована”, translated as “consistent”.

Original source details: GELFAND-P001265

$\Rightarrow$ it is believed that this is a unified field theory (i.e. space is 10-dimensional, there is super., etc.)

GELFAND-P001266

[source number: 15]

(8) Into field theory: a string is an infinite number of particles (harmonics). This means that ordinary field theory embeds here.



Diagram and typed description.

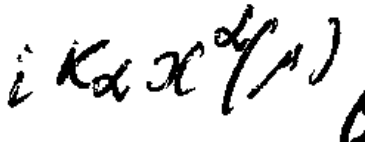


Original source details: GELFAND-P001266

$$\int_p Dx\, Dg\, e^{-\int \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu d^2\xi} \prod_\alpha \int e^{ik_\alpha x^\mu(\xi_\alpha)} d^2\xi_\alpha \sim$$

$$\sim \int_{\substack{M_p \\ y_1,\dots,y_{3p-3}}} \frac{|d\mu_p|^2}{|\det \operatorname{Im} T|^{13}} \underbrace{K(k_\alpha, y, \bar{y})}_{\substack{\| \\ \prod_\alpha \int d^2\xi_\alpha \prod_{\alpha,\beta} e^{k_\alpha k_\beta G_R(\xi_\alpha, \xi_\beta | y, \bar{y})}}} \tag{9}$$

Uncertain reading. Argument in the exponent on the first line: ξ_α?



Original source details: GELFAND-P001266

$\sum k_\alpha = 0$,

G_R — rel. Green’s function of the surface.

Uncertain reading. Abbreviation before ‘Green’s function’: «рел.» or «рег.»

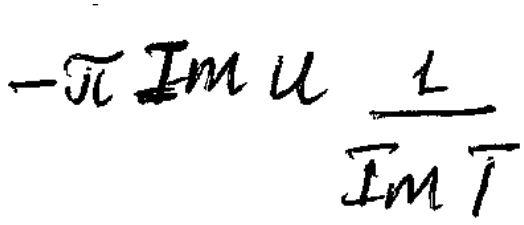


Original source details: GELFAND-P001266

$$\exp G(\xi_\alpha, \xi_\beta) \sim e^{-\pi \operatorname{Im} u \frac{1}{\operatorname{Im} T}} |\Theta(u, T)|^2, \tag{10}$$

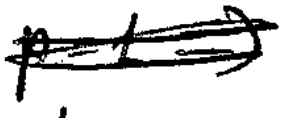
$u_i = \int_{\xi_\alpha}^{\xi_\beta} \omega_i$ | T — period matrix of the given curve;
$d\mu_p$ — a certain holomorphic measure on the moduli space

(see last time)



Uncertain reading. One factor of $\operatorname{Im} u$ is visible in the exponent.

Original source details: GELFAND-P001266



Uncertain reading. Crossed out: $p = 1 \Rightarrow$?

Original source details: GELFAND-P001266

$d\mu_p$ has been computed for $p \leq 4$, unknown beyond that (only Manin's less explicit formulas)

NB: a way of writing Green's function (of the Bergman metric) on a curve! — by means of a map into the Jacobian



Undeciphered passage. Crossed-out word after 'Jacobian'

Original source details: GELFAND-P001266

(Jacobi) and then use Θ-functions.

On the other hand

$$d\mu_p = \prod_i dy_i \frac{\det' \Delta_{-1}}{(\det' \Delta_0)^{13}} \tag{11}$$

$$\delta_{\eta_\alpha} \ln \det' \Delta_0 = \operatorname{Tr}(\partial \eta_\alpha)(\partial G) \stackrel{?}{=} \partial_\xi \eta_\alpha \partial_\xi G(\xi, \xi')\big|_{\xi'=\xi} \tag{12}$$

A full investigation has not been carried out.

$$\partial^2 G(\xi, \xi')\big|_{\xi'=\xi} = \frac{\Theta_{,i}\omega_i'' + \Theta_{,ijk}\omega_i\omega_j\omega_k}{\Theta_{,i}\omega_i} - \frac{3}{2}\left(\frac{\Theta_{,i}\omega_i'}{\Theta_{,i}\omega_i}\right)^2, \tag{13}$$

ω' — the derivative of ω with respect to ξ.

GELFAND-P001267

The realistic theory is the superstring. The impression is that the standard notion of superspace is not quite adequate.

Stat. sum after integration over the odd moduli?



Uncertain reading. The initial abbreviation reads «Стат.» (‘stat.’)

Original source details: GELFAND-P001267

It should be

$$\int_{M_p} \frac{|d\mu_p^{SS}|^2}{(\det \operatorname{Im} T)^5} \tag{14}$$

(Under modular transformations $T \to (AT+B)/(CT+D)$

$$\frac{d\mu_p^{SS}}{d\mu_p^{bos}} = f_8 \qquad \text{— a form of weight 8)} \tag{15}$$

It would be good if there were no cusp forms of weight 8

Genus $1, 2, 3$ — there are none.

Genus 4 — there are some

GELFAND-P001268

10/III–86 Gelfand Sem.

A. Morozov Low-energy limit of string theory

Gauge-theory Lagrangian

$$\operatorname{Tr} F_{\mu\nu}^2 \tag{1}$$

$$F_\mu^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + i \underbrace{f^{abc}}_{\text{structure const.}} A_\mu^b A_\nu^c \tag{2}$$

or

$$F = dA + [A, A] \tag{2'}$$

Gauge transformations:

$$A_\mu \to A_\mu + [A_\mu, \varepsilon] + \partial_\mu \varepsilon \tag{3}$$

How can we embed this in string theory? We flatten the strings into pointlike objects

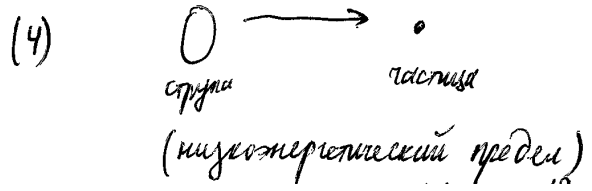


Diagram and typed description.

(4) string → particle

(low-energy limit)

Original source details: GELFAND-P001268

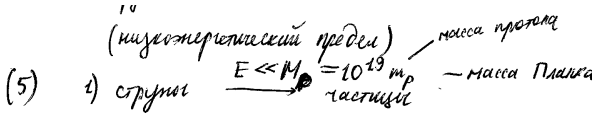


Diagram and typed description.

(5) 1) strings $\xrightarrow{E \ll M_P = 10^{19} m_p}$ particles (proton mass)

Original source details: GELFAND-P001268

— Planck mass

Undeciphered passage. Crossed-out passage



Original source details: GELFAND-P001268

($M_P^2 = \overline{h}c/\gamma$, γ — grav. const.)

2) particles different fields
different masses

3) massless vectors (particles of mass 0) are described precisely by Lagrangian (1) (this follows from string dynamics)

Where does the gauge group in string theory come from? We have to make the string live

Undeciphered passage. Crossed-out passage



Original source details: GELFAND-P001268

in non-flat space/time.

GELFAND-P001269

[source number: 18]

I. If $x^\mu(\xi) : C \to \mathbb{R}^{26}$ (flat), then vectors without the index i (in $F^i_{\mu\nu}$) $\Rightarrow$ abelian group $U(1)$

$$\int d^2\xi \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu \tag{A}$$

II. The method of Chan–Paton gives $\forall$ group.

III.

Uncertain reading. Crossed-out mark: ‘A’?



Original source details: GELFAND-P001269

Consider $C \to T^k \otimes \mathbb{R}^{26-k}$

In the interesting case $k = 16 \rightsquigarrow$ a superstring from a bosonic string.

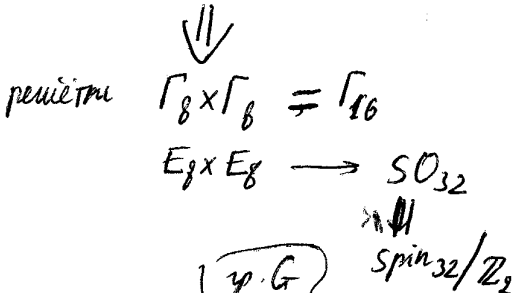


Diagram and typed description.

lattices $\Gamma_8 \times \Gamma_8 \quad = \quad \Gamma_{16}$

$E_8 \times E_8 \longrightarrow SO_{32}$

$Spin_{32}/\mathbb{Z}_2$

Original source details: GELFAND-P001269



Original source details: GELFAND-P001269

Uncertain reading. The sign between the lattices reads $\neq$?



Original source details: GELFAND-P001269

Uncertain reading. Two marks to the left of the arrow toward $Spin_{32}/\mathbb{Z}_2$.

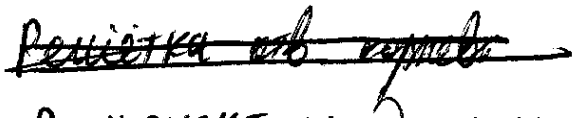


Original source details: GELFAND-P001269

Undeciphered passage. Marks at the upper right of the lattice diagram

gp. G

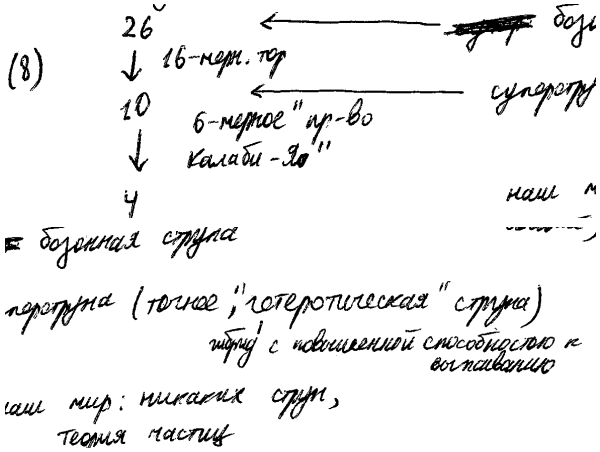
(6) T^k from the lattice of root vectors of the Lie alg. $\Rightarrow$ one gets the corresponding Yang–Mills theory with gauge group G



Original source details: GELFAND-P001269

Uncertain reading. Crossed out: ‘Lattice [of?] roots’?

(7) At the birth of the world it was ∞-dimensional, then compactified into a thin film around 4-dimensional space (thickness $\sim 10^{-33}$ cm; the proton’s size is 10^{19} times larger).



Original source details: GELFAND-P001269

Diagram and typed description.

(8)

26 ← bosonic string

16-dimensional torus

10 ← superstring (more precisely, ‘heterotic’ string)

6-dimensional ‘Calabi–Yau space’

a hybrid with enhanced ability to survive

4 our world: no strings, particle theory



Original source details: GELFAND-P001269

Undeciphered passage. Crossed-out marks before ‘bosonic string’

(9) Question: according to what eqs. does compactification occur?

In diagram (8) the odd variables are left out (the dimensions are even)

Read: Kac, Frenkel — Invent. or Adv. in Math. (on the duality of bosonic and fermionic models)

Physics papers: Freund P.

Gross, Harvey, Martinez, Rohm Nucl. Phys. B, 1985.

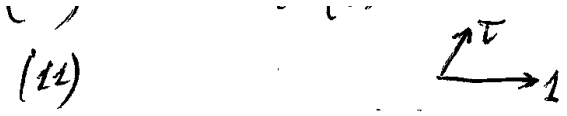


Original source details: GELFAND-P001269

Uncertain reading. The manuscript reads 'Martinez'.

Consider the map

$$x^\mu(\xi) : T^2 \to T^{16} \times \mathbb{R}^{10} \tag{10}$$



Original source details: GELFAND-P001269

Diagram and typed description.

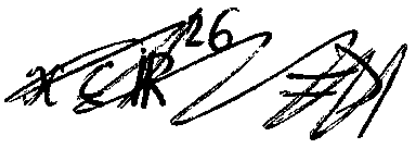
(11) $\tau \to 1$

— the lattice defining T^2 (one-loop string diagrams)

GELFAND-P001270

We shall compute

$$Z = \int Dx\, Dg_{ab}\ \exp \int d^2\xi \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu \tag{12}$$



Original source details: GELFAND-P001270

Uncertain reading. Crossed out: $x \in \mathbb{R}^{26} \Rightarrow$

$x \in T^{16} \times \mathbb{R}^{10} \Rightarrow$

$$\left.\begin{cases} x^\mu(\xi + 1) = x^\mu(\xi) + \Lambda_2^\mu \\ x^\mu(\xi + \tau) = x^\mu(\xi) + \Lambda_1^\mu \end{cases}\right| \tag{13}$$



Original source details: GELFAND-P001270

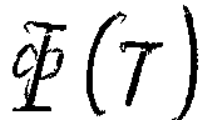
Uncertain reading. In the first line the sign before Λ_2^μ is heavily overwritten. $+$?

$$\Lambda_1^\mu = 0, \quad \mu = 1, \dots, 10; \quad \Lambda_1^\mu \in \text{the lattice for } \mu = 11, \dots, 26. \tag{14}$$

The answers obtained:

$$\underset{\Gamma_{16}}{\text{for}} \qquad \int_{M_p} Z_{[C \to T^{16} \times \mathbb{R}^{10}]} = \int_{M_p} Z_{[C \to \mathbb{R}^{26}]} \cdot \Phi(T) \tag{15}$$

Uncertain reading. One argument T is written in (15).



Original source details: GELFAND-P001270

$$\Phi(T, \bar{T}) = \sum_{\substack{\text{even} \\ \text{charact. } m}} |\Theta_m^{16}(T)|^2 \tag{16}$$

$$F(T) = \sum_{\substack{\text{even} \\ \text{charact.}}} \Theta_m^{16}(T) \tag{17}$$

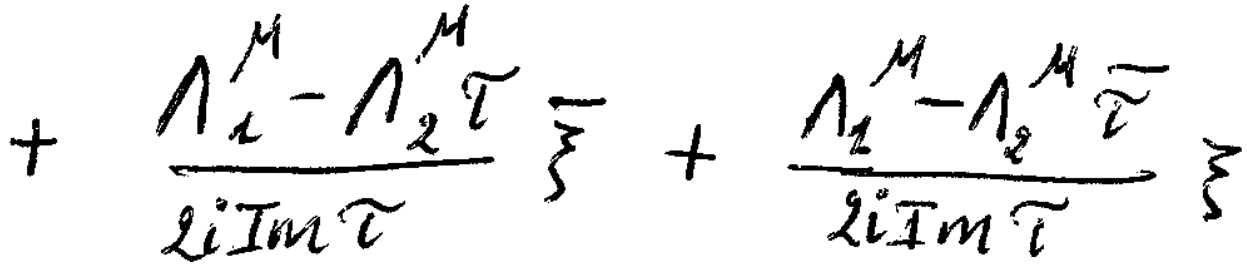
Let

$$x^\mu(\xi) = \tilde{x}^\mu(\xi) + \frac{\Lambda_1^\mu - \Lambda_2^\mu \tau}{2i \operatorname{Im} \tau} \bar{\xi} + \frac{\Lambda_1^\mu - \Lambda_2^\mu \bar{\tau}}{2i \operatorname{Im} \tau} \xi \tag{18}$$

Uncertain reading. A $+$ is written before the second fraction term.



Original source details: GELFAND-P001270



Original source details: GELFAND-P001270

Uncertain reading. Both signs between the summands are $+$.

Then

(19) $\tilde{x}^\mu$ satisfies (13) without Λ_j^μ (as for a map into $\mathbb{R}^{26}$).

Integral (12) after integrating over x and g becomes

$$\int_M Z[T^2 \to T^{16} \times \mathbb{R}^{10}] \tag{20}$$



Uncertain reading. The sign $\times$ is crossed out in (20)

Original source details: GELFAND-P001270

(M — moduli space of Riemann surfaces)

$$Z[T^2 \to T^{16} \times \mathbb{R}^{10}] = \sum_{\Lambda_1, \Lambda_2} Z[T^2 \to \mathbb{R}^{26}] \exp\left(-\frac{|\Lambda_1 - \Lambda_2\tau|^2}{\operatorname{Im}\tau}\right) \tag{21}$$



Original source details: GELFAND-P001270

Uncertain reading. The sign $\times$ is crossed out in (21)



Original source details: GELFAND-P001270

Uncertain reading. The zero in the exponent 10 is circled or overwritten.

Integration over x (det), over $g \to e^{\varphi}\tilde{g}$

in dim. 26 the dependence on φ disappears etc.

GELFAND-P001271

[source number: 19]

Naively there is no dependence on φ since the dimension $= 2$. (The contributions of $\sqrt{g}$ and g^{ab} cancel)

So we need to know:

$$\Phi(\tau, \bar{\tau}) = \sum_{\Lambda_1, \Lambda_2 \in \Lambda} \exp -\frac{|\Lambda_1 - \Lambda_2\tau|^2}{\operatorname{Im}\tau} \tag{22}$$

Λ — the lattice corresponding to the 16-dimensional torus



Original source details: GELFAND-P001271

Uncertain reading. Crossed out: ‘чтоб’ (‘so that’)?

(does not possess any properties, but sometimes)

Generally speaking, it reduces to a finite sum of squared absolute values of anal. functions.

$$\varphi(\tau,\bar{\tau}) = \sum_{i=1}^{n} |f_i(\tau)|^2, \qquad f_i \text{ — analytic functions.} \tag{23}$$

There are 2 examples

This happens when



Original source details: GELFAND-P001271

Uncertain reading. The abbreviation before ‘when’ reads ‘быв.’.

(24)



Original source details: GELFAND-P001271

Uncertain reading. Crossed-out mark before ‘lattice’: ‘2)’?

the lattice is even and self-dual

For Γ_{16}, the formula apparently holds

$$\varphi(\tau,\bar{\tau}) = |\Theta_{01}^{16}(\tau)|^2 + |\Theta_{10}^{16}(\tau)|^2 + |\Theta_{00}^{16}(\tau)|^2 \tag{25}$$

(Apparently evenness is unnecessary)

Also put

$$F(\tau) \equiv \sum_{i=1}^{n} f_i(\tau) \tag{26}$$

Let us require that F behave well. Then the superstring and heterotic string are defined. It is interesting whether anything remains for lattices that are not self-dual. It turns out that

$$F(\tau) = \sum_{\lambda\in\Lambda} e^{i\pi\lambda^2\tau} \equiv \Theta \tag{27}$$

Next, we obtain the measures

$$|d\mu^{SS}(y)|^2 = |d\mu^{bos}(y) f^{ferm}(y)|^2 \tag{28}$$

(superstr.)



Original source details: GELFAND-P001271

Uncertain reading. After $d\mu^{bos}(y)$, $|^2$ is crossed out.

$$d\mu^{het}(y,\bar{y}) = |d\mu^{bos}(y)|^2 f^{ferm}(y)\bar{F}(\bar{y}) \tag{29}$$

Here

$$|d\mu^{bos}(y)|^2 = Z[T^2 \to \mathbb{R}^{26}] \tag{30}$$

(31) For even self-dual lattices the sum (26) transforms as a modular form



Uncertain reading. The end of the word ‘self-dual’ is overwritten.

Original source details: GELFAND-P001271

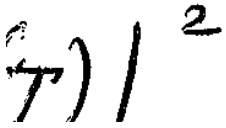
(32) It is unclear whether that holds only for even self-dual ones

GELFAND-P001272

(33) In $\mathbb{R}^{16}$ there are only 2 even self-dual lattices. (see Serre, ‘A Course in Arithmetic’)
In terms of $F(\tau)$ these lattices are indistinguishable

$$F_{\Gamma_{16}}(T) = \sum_{\substack{\text{even} \\ m}} \Theta_m^{16}(T), \tag{34}$$

$$F_{\Gamma_8 \times \Gamma_8}(T) = \left| \sum_{\substack{\text{even} \\ m}} \Theta_m^{8}(T) \right|^2. \tag{35}$$



Uncertain reading. In (35) the vertical bars and square are written.

Original source details: GELFAND-P001272

We want:

$$\text{Strings} \to \text{particles} \to \text{massless particles}. \tag{36}$$

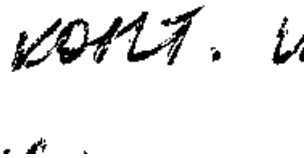
1 More general integrals are needed:



Undeciphered passage. Crossed-out writing before ‘Vertex’

Original source details: GELFAND-P001272

Vertex opers. — under the cont. integral



Uncertain reading. The abbreviation before ‘integral’ is ‘конт.’ (‘cont.’).

Original source details: GELFAND-P001272

$$V(k_\alpha) = \int d^2\xi \, e^{ik^\mu x^\mu(\xi)} \tag{37}$$

$$V^\mu(k_\alpha) = \int d^2\xi \, \partial x^\mu(\xi) e^{ik^\mu x^\mu(\xi)} \tag{38}$$

(39)

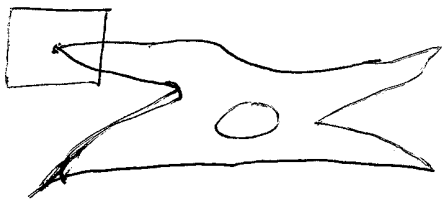
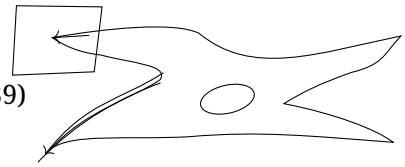


Diagram and typed description.

(39)



Original source details: GELFAND-P001272

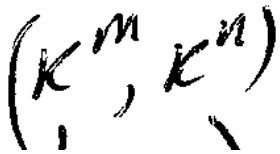
Weights can be added in the frames

Consider $T^{16} \times \mathbb{R}^{10}$ instead of $\mathbb{R}^{26}$.

Take the vertices corresponding to $\mathbb{R}^{10}$.

$$V^m(k) = \int d^2\xi \, \partial x^m e^{ik^m x^m} \cdot \underbrace{e^{ik^M x^M}}_{\substack{\text{This distinguishes it} \\ \text{from the 26-dimensional case}}} \tag{40}$$

[Under k:] $\left(\underset{\text{10-dimensional}}{k^m}, \underset{\text{16-dimensional}}{k^M}\right)$



Uncertain reading. The superscript of the second component reads M?

Original source details: GELFAND-P001272

We need to know [inserted: new] correlators. This leads to commut. relations,

$$[e^{ik^M x^M}, e^{ik'^M x^M}] \sim e^{i(k^M + k'^M)x^M}, \tag{41}$$

if $k^2 = k'^2 = 2$, $(k + k')^2 = 2$, and 0 in the remaining cases.

(Kac, Frenkel)

GELFAND-P001273

[source number: 20]

the commutators are understood as follows

$$x = \sum a_n z^n + \ldots \qquad \frac{\partial}{\partial z} \tag{42}$$

This leads to the appearance of groups in the string:

$$E_8 \times E_8 \quad \text{or} \quad SO_{32}$$

GELFAND-P001274

24.III.86. Gelfand Sem.



Original source details: GELFAND-P001274

Uncertain reading. The manuscript reads ‘24.III.86.’. Catalogue record 672 dates the talk 24 March 1987.

Fuchs. Cohomology

$\mathbb{R}^n \setminus \bigcup L_i$, L_i — (affine) planes

$\mathbb{R}^n \supset \bigcup L_i$, L_i — affine subspaces of arbitrary dimensions.

$$X = \mathbb{R}^n \setminus \bigcup L_i.$$

How to calculate $H_k(X)$

Suppose L_i — hyperplanes

Then $H_k = 0$ for $k \neq 0$.

Answer (Zaslawsky, 1975)

Construct



Original source details: GELFAND-P001274

Uncertain reading. Crossed out: possibly ‘such a’.

partially ordered set



Original source details: GELFAND-P001274

Undeciphered passage. Crossed-out writing before A

A with elements $\mathbb{R}^n$, L_i and all their nonempty intersections



Original source details: GELFAND-P001274

Uncertain reading. The endings in the source read ‘непустым пересечения’.

Möbius function

$$\mu(\mathbb{R}^n) = 1, \qquad \mu(L_i) = -1$$

$$\mu(v) = -\sum_{w>v} \mu(w)$$

$$h_0 = \sum_{v \in A} |\mu(v)| \qquad \leftarrow \text{Zaslavsky's formula.}$$



Uncertain reading. The letter A under the sum is overwritten.

Original source details: GELFAND-P001274

Example.

In $\mathbb{R}^3$ coordinate planes

$\mathbb{R}^3$,	L_x,	L_y,	L_z,	L_{xy},	L_{xz},	L_{yz},	0
1	-1	-1	-1	1	1	1	-1



Uncertain reading. The last letter in L_{yz} is overwritten.

Original source details: GELFAND-P001274

$$\sum_{1}^{8} |\pm 1| = 8, \quad \text{as expected (8 octants)}$$

Main case (Goresky).

A

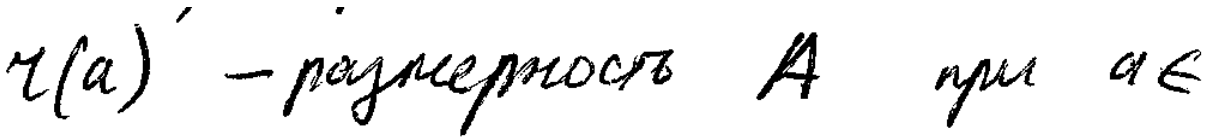


Undeciphered passage. Crossed-out writing after A

Original source details: GELFAND-P001274

is defined in the same way; A — partially ordered set

$\tau(a)$ — dimension of A for $a \in$



Original source details: GELFAND-P001274

Uncertain reading. The first symbol reads τ or r. The writing stops after $a \in$.

Theorem.

$$H_i(X) = \bigoplus_{a \in A} H_i(a),$$

(Goresky–MacPherson)

where

$$H_i(a) = H^{n-\tau(a)-i-1}\big(X_{>a}, X_{(a,\mathbb{R}^n)}\big),$$



Original source details: GELFAND-P001274

Uncertain reading. The letter in the exponent reads τ or r.



Original source details: GELFAND-P001274

Uncertain reading. The second X is overwritten. Its earlier form is unclear.

GELFAND-P001275

[source number: 29]

where

$$\left.\begin{matrix} > a \\ (a, \mathbb{R}^n) \end{matrix}\right\} \text{ subsets of the set } A \quad \begin{matrix} \leftarrow \{b \mid b > a\} \\ \leftarrow \{b \mid b > a\} - \{\mathbb{R}^n\} \end{matrix}$$

X_M, where M — a partially ordered set (in particular, $M \subset A$) is by definition a simplicial complex with vertices the elements of M, and p-dimensional simplices the subsets

$$\{a_0 < \cdots < a_p\} \subset M$$

(Comment: by definition we take

$$H^i(\varnothing, \varnothing) = \begin{cases} \mathbb{Z}, & i = -1, \\ 0, & i \neq -1. \end{cases}$$

The pair $(X_{>a}, X_{(a,\mathbb{R}^n)}) = (\varnothing, \varnothing)$ only if $a = \mathbb{R}^n$.

Indeed, $X_{>a}$ is always contractible, so $H^k(X_{>a}, X_{(a,\mathbb{R}^n)})$ reduces to the cohomology of $X_{(a,\mathbb{R}^n)}$, namely

$$H^i\left(\begin{matrix}\text{contractible} \\ \text{space}\end{matrix}, B\right) = \begin{cases} H^{i-1}(B), & i > 1 \\ H^0(B)/\mathbb{Z}, & i = 1. \end{cases}$$

In the Goresky



Original source details: GELFAND-P001275

Uncertain reading. Crossed out, possibly ‘and’.

— MacPherson theorem the coefficients are arbitrary.

Special case: Orlik–Solomon thm. (



Uncertain reading. An overwritten sign before ‘when’, possibly ‘в’.

Original source details: GELFAND-P001275

when everything is complex).

Example.

$\mathbb{R}^3 \setminus$ coordinate planes and one more line through 0, but lying in none of the hyperplanes.

$\underline{8}$ — 0-dimensional and $\underline{2}$ 1-dimensional.

Question: given an arbitrary partially ordered subset, what is the meaning of what is on the left (something like a toric variety

(I.M.G.)



Uncertain reading. Abbreviated ‘мн-зия’ or ‘м-зия’ in the source.

Original source details: GELFAND-P001275

)

On the proof.

Consider a point [inserted above:] x_0 in general position in $\mathbb{R}^n$ and a ball centred at x_0; we shall expand it, tracking changes in the homology of what is inside (need relative homology).



Uncertain reading. The insertion reads x_0 or a_0.

Original source details: GELFAND-P001275

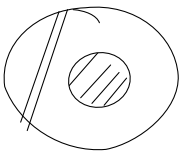


Uncertain reading. After ‘need’ the visible letters read ‘на’, perhaps intended as ‘нам’ (‘we need’).

Original source details: GELFAND-P001275



Diagram and typed description.



Original source details: GELFAND-P001275

ball with a line removed

mod the excised smaller ball.
GELFAND-P001276

In fact the problem does not reduce to computing relative homology.



Uncertain reading. Crossed out: D_{small} and D_{large} with a sign that could be $\subset$ or $\supset$.

Original source details: GELFAND-P001276

$$D_{\text{small}} \subset D_{\text{large}}$$

$$H_*(D_{\text{large}}) = H_*(D_{\text{small}}) \oplus H_*(D_{\text{large}}, D_{\text{small}})$$



Uncertain reading. The end of the subscript 'large' in the relative group is overwritten.

Original source details: GELFAND-P001276

if the exact sequence splits, but here this is checked by purely geometric means.
(1)



Uncertain reading. Crossed out: 'When'.

Original source details: GELFAND-P001276

Example, where it is carried through to the end.

$$\operatorname{codim} L_i = k \qquad \text{(independent of } i\text{)}$$
$$\operatorname{codim}(L_{i_1} \cap \cdots \cap L_{i_q}) \quad \text{is divisible by } k$$

($k = 1$ is satisfied, and in the complex case it is satisfied)
But even



Uncertain reading. Crossed out, possibly 'with'.

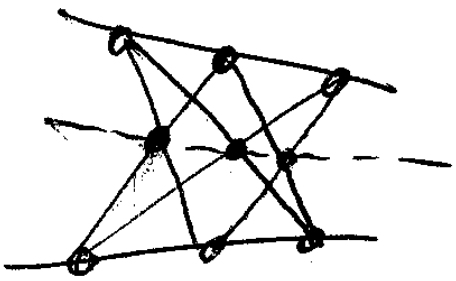
Original source details: GELFAND-P001276

in $\mathbb{R}^4$ with $k = 2$ this need not be a complex situation!! (There is an example of Goresky's)

Question (ИМГ): under what conditions does there exist a complex structure in which they all become complex?

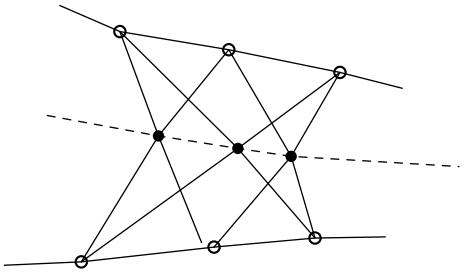
Goresky's example.

Pappus's thm..



Original source details: GELFAND-P001276

Diagram and typed description.

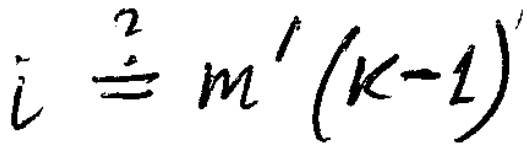


Under complex perturbation the situation persists, but under real perturbation it does not!!

Suppose the conditions of (1). hold. Then

$$\forall a,\ H_i(a) \neq 0 \quad \text{only for one } i:$$

$$i \stackrel{?}{=} m'(k-1)$$



Uncertain reading. A question mark stands above the equality. There is a faint stroke after the parenthesis, possibly another prime.

Original source details: GELFAND-P001276

$$H_i(a) = \underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{|\mu(a)|}$$

Question of I.M.G.. How does one find the right basis for cohomology?

GELFAND-P001277

[source number: 30]



Undeciphered passage. Crossed-out writing before the name

Original source details: GELFAND-P001277

Varchenko. Mixed Hodge structures

These are 2 filtrations in a vector space, increasing and decreasing

(in the cohomology of a complex variety) Hodge and weight

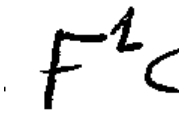
[Below 'weight', by the stroke:] topological (unchanged under deformation of the complex structure)

They are suitably compatible and vary suitably when the variety changes. There will be an example of this.

Cohomology of a nonsingular projective variety (compact)

In cohomology there is a pure Hodge structure

$$0 \subset F^k \subset \cdots \subset F^1 \subset F^0 = H^k \tag{1}$$



Uncertain reading. The superscript reads 1. The upper horizontal stroke joins F.

Original source details: GELFAND-P001277

Here

$$F^s = \{[\omega], \quad \omega \text{ — closed complex-valued differential form,} \\ \omega = \sum a_i \underbrace{dz_{i_1} \wedge \cdots \wedge dz_{i_s}}_{\text{in all } k} \wedge \underbrace{\cdots}_{}\} \tag{2}$$

$\exists$ a representative: ($\geq s$ holomorphic differentials in the local expression)

Change of complex structure $\Rightarrow$ the subspaces vary continuously.

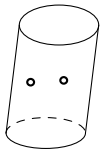
Example of a weight filtration.

In $\mathbb{C} \setminus \{0\}$ identify 2 points $*, *$

Topologically:



Diagram and typed description.



Original source details: GELFAND-P001277

$$\dim H^1(X) = 2$$

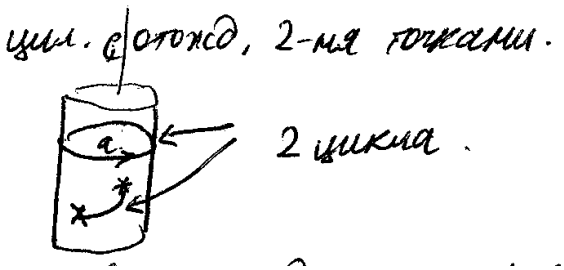
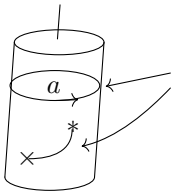


Diagram and typed description.

cyl. with 2 points identified.



2 cycles.

Original source details: GELFAND-P001277

In H_1 there is a 1-dimensional subspace generated by the cycle a



Uncertain reading. The superscript is crossed out, possibly 1.

Original source details: GELFAND-P001277

$W \subset H^1(X)$



Undeciphered passage. Crossed-out writing after W

Original source details: GELFAND-P001277

[Below the cancellation, by the oblique stroke:] the annihilator of this 1-dimensional subspace.

[By the vertical stroke under W:] weight space.

Another way mixed Hodge structures arise connected with asymptotics.

Consider a real-analytic function $f : \mathbb{R}^n \underset{\text{analytic}}{\longrightarrow} \mathbb{R}$.

$\omega = h\, dx_1 \wedge \cdots \wedge dx_n$ with compact supp

$$\omega \mapsto \int_{\mathbb{R}^n} f^\lambda \omega \tag{3}$$

extends



Uncertain reading. Crossed out before ‘на’, possibly ‘на’ or ‘не’.

Original source details: GELFAND-P001277

to all of $\mathbb{C}$ with poles at the points

$$\int_{\mathbb{R}^n} f^\lambda \omega = \sum a_{k,\alpha}(\omega) \frac{1}{(\lambda+\alpha)^k} \tag{4}$$



Uncertain reading. A crossed-out mark to the right of (4), possibly α or a.

Original source details: GELFAND-P001277

GELFAND-P001278

k — a manifestation of the weight filtration



Original source details: GELFAND-P001278

Uncertain reading. Crossed out between ‘weight’ and ‘filtration’: ‘Hodge str.’.

α —”— Hodge —”—

2 parameters, the integer k and the rational α are always such a manifestation of a mixed Hodge str..



Original source details: GELFAND-P001278

Uncertain reading. The manuscript reads ‘таким’ (‘such a’).

Complex case.

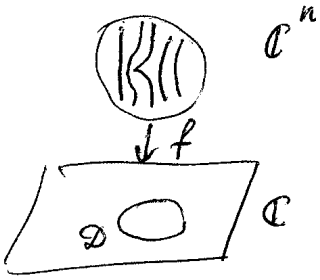
$f : (\mathbb{C}^n, 0) \longrightarrow (\mathbb{C}, 0), \quad f$ holom..

0 — an isolated critical point

Consider the fibre $f^{-1}(0)$ — singular, with isolated singular point 0.

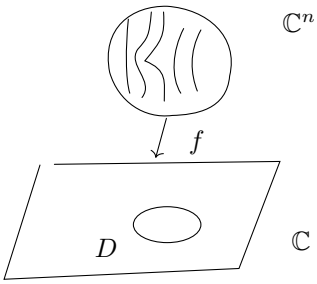
Consider in $\mathbb{C}^n$ a small ball and in $\mathbb{C}$ a small disc (much smaller) $\Rightarrow$

$\Rightarrow$ one gets holomorphic fibres (not algebraic at all)



Original source details: GELFAND-P001278

Diagram and typed description.

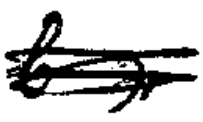


$X_t = f^{-1}(t)$ — one fibre.



Original source details: GELFAND-P001278

Uncertain reading. The X in X_t is overwritten.



Original source details: GELFAND-P001278

Uncertain reading. Crossed out after $f^{-1}(t)$, possibly ‘in...’.

$(n-1)$-dimensional complex variety.

Milnor's thm.:

$$X_t \underset{\text{homotopy equiv.}}{\sim} \underbrace{S^{n-1} \vee \cdots \vee S^{n-1}}_{\substack{\mu \text{ — the Milnor number} \\ \text{of the critical point}}}$$

$n = 2 \Rightarrow X_t$ — a noncompact complex curve, connected

At the boundary something is cut off

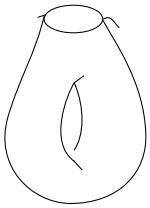


Uncertain reading. Crossed out after 'boundary', possibly 'of the fibre'.

Original source details: GELFAND-P001278

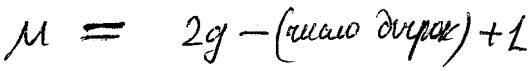


Diagram and typed description.



Original source details: GELFAND-P001278

$$\mu = 2g - (\text{number of holes}) + 1 \tag{5}$$



Uncertain reading. In (5) a minus is visible before the parentheses and $+1$ after them.

Original source details: GELFAND-P001278



Diagram and typed description.



Original source details: GELFAND-P001278

One gets a bouquet of circles.

Going around produces the monodromy transformation $M : H^{n-1}(X_t) \longrightarrow H^{n-1}(X_t)$

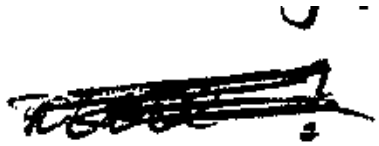


Uncertain reading. Crossed-out heading, possibly 'Integrals'.

Original source details: GELFAND-P001278

Gelfand's question: what about $\int f_1^{\lambda_1} f_2^{\lambda_2} \cdots f_k^{\lambda_k}$ — ?

For $\mathbb{C}^n \to \mathbb{C}^k$ will something like a mixed Hodge str. arise?



Undeciphered passage. Crossed-out writing between 'will' and 'arise'

Original source details: GELFAND-P001278

I.M.G.'s question: what arises for a mapping given by a pair of quadratic forms?



Uncertain reading. Crossed out after 'what', possibly 'бы'.

Original source details: GELFAND-P001278

Consider the form

$\omega = h\, dz_1 \wedge \cdots \wedge dz_n, \qquad h$ — holom..

On a nonsingular fibre one can consider

ω/df — holom. $(n-1)$-form

Take a distinguished cycle in the fibre and extend it continuously to nearby fibres. We obtain a multivalued family of cycles $\delta(t)$



Uncertain reading. A short rising stroke after $\delta(t)$, possibly a comma.

Original source details: GELFAND-P001278

GELFAND-P001279

[source number: 31]

$$I(t) = \int_{\delta(t)} \frac{\omega}{df} \tag{6}$$

Passage through a critical value — Morse's thm.

going around 0 — Picard–Lefschetz thm.

These 2 aspects give 2 filtrations.

Weight filtration — a filtration of the linear operator M by its Jordan blocks.



Uncertain reading. A short overwritten mark over the end of 'Весовая' ('Weight'), possibly 'й'.

Original source details: GELFAND-P001279



Undeciphered passage. Crossed-out writing before ‘linear operator’

Original source details: GELFAND-P001279

For M we need to construct a nilpotent matrix and its filtration

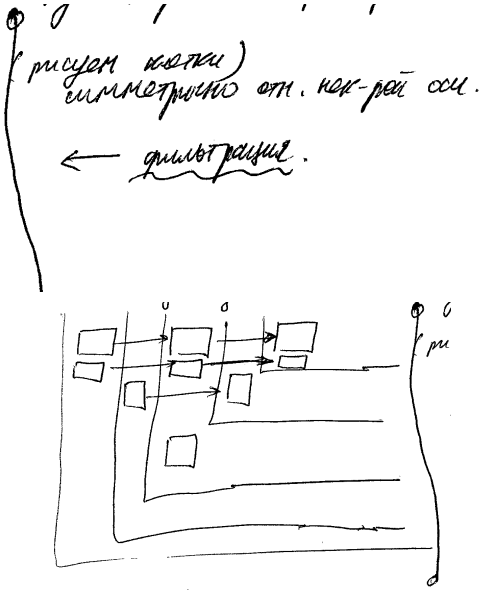
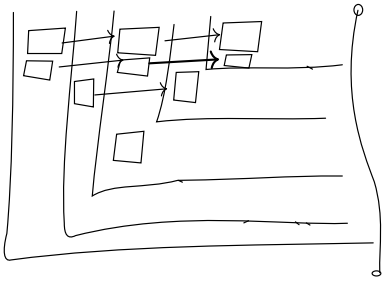


Diagram and typed description.



(draw the blocks) symmetrically about some axis.
⟵ filtration.

Original source details: GELFAND-P001279

1) Eigenvectors in the largest Jordan blocks

2) Add eigenvectors in smaller blocks



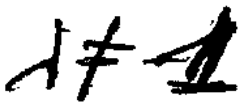
Undeciphered passage. Crossed out after ‘blocks’

Original source details: GELFAND-P001279

and so on

Different eigenvalues are grouped depending on whether

$$\underset{\text{middle weight } n}{\lambda = 1} \quad \text{or} \quad \underset{\text{weight } n-1 \text{ for all}}{\lambda \neq 1}$$



Uncertain reading. The 1 in $\lambda \neq 1$ is overwritten.

Original source details: GELFAND-P001279

$$I(t) = \int_{\delta(t)} \frac{\omega}{df} = \sum a_{k,\alpha}\, t^{\alpha} (\ln t)^k \tag{7}$$



Uncertain reading. Crossed out before $a_{k,\alpha}$, possibly ‘where’.

Original source details: GELFAND-P001279

$a_{k,\alpha}$ depend linearly on ω and on δ

$e^{2\pi i\alpha}$ = eigenvalues of M

k = [inserted above: sizes] of Jordan blocks of the operator M



Uncertain reading. The insertion reads ‘sizes’ or ‘size’.

Original source details: GELFAND-P001279

References:

1) Varchenko, DAN, 1980

2) the construction in VINITI

Undeciphered passage. Crossed-out mark before ‘the construction’



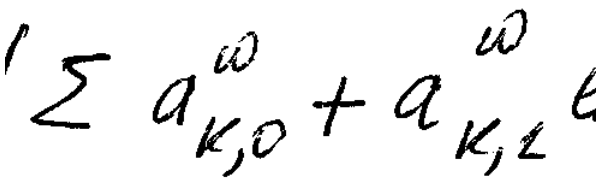
Original source details: GELFAND-P001279

3) Book on singularities

GELFAND-P001280

Leading term of the asymptotics.

$$I(t) = t^{\alpha(\omega)} \left(\sum a^{\omega}_{k,0} + a^{\omega}_{k,1} \ln t + \dots \right) + o\left(t^{\alpha(\omega)} \right) \tag{8}$$



Uncertain reading. The sum sign in (8) has no index. The coefficients show subscripts $k, 0$ and $k, 1$.

Original source details: GELFAND-P001280

The mixed Hodge str. is obtained from the first term.
The Bernstein polynomial comes from all the terms.

$V_\alpha = \{$classes of the leading parts of forms $\omega,\ \alpha(\omega) = \alpha\}$ — cohomology classes (α — rational)

$$V_\alpha \subset V_{\alpha+1}$$

Varchenko was interested in the semicontinuity of the leading parts of the asymptotics.

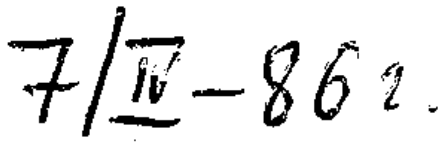

полунепр.

Uncertain reading. The abbreviation is ‘полунепр.’ (‘semicontinuity’).

Original source details: GELFAND-P001280

GELFAND-P001281

7/IV—86 Gelfand sem.


7/IV—86 г.

Uncertain reading. The handwritten year is ‘86’.

Original source details: GELFAND-P001281

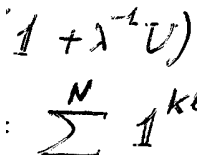
Cherednik. Yangian.
Definition of the Yangian.

$$\underset{\text{matr. } N\times N}{M_N} \supset \{\mathbf{1}^{k\ell}\}, \qquad \mathbf{1} \text{ — matrix identity}$$

[Slanting line from $\{\mathbf{1}^{k\ell}\}$ to ‘std. basis’.] (1 in position k, ℓ, the others are 0)

$$(\mathbf{1}^{k\ell})_{ij} = \delta_{ik}\delta_{j\ell}.$$

$$\det(\mathbf{1} + \lambda^{-1}U) \text{ — ?}$$


1 + λ⁻¹U)
: ∑ 1^k

Uncertain reading. The letter in the determinant reads as U.

Original source details: GELFAND-P001281

$$U = \sum_{k,\ell=1}^{N} \mathbf{1}^{k\ell} e_{\ell k} \in M_N\big(U(g\ell_N)\big)$$

$$\{e_{\ell k}\} \subset U(g\ell_N)$$



Uncertain reading. The initial letter of $U(g\ell_N)$ has the same form as the one in the determinant.

Original source details: GELFAND-P001281

(1) [Crossed out]



Uncertain reading. The strokes appear to cover $L_n(\lambda)$.

Original source details: GELFAND-P001281

$$L_n(\lambda) = E\mathbf{1} + \sum_{k,\ell} \sum_{q=1}^{n} \frac{1}{\lambda^q} \mathbf{1}^{k\ell} E_{\ell k}^{q-1} =$$

coeff. are the letters $E_{\ell k}^{q}$ (with 3 indices)

$$= \begin{pmatrix} E & & & 0 \\ & E & & \\ & & \ddots & \\ 0 & & & E \end{pmatrix} + \cdots$$

[At right:] (the index n is sometimes omitted below)

Set

$$R(\lambda) := \mathbf{1} \otimes \mathbf{1} + \frac{1}{\lambda} \sum \mathbf{1}^{k\ell} \otimes \mathbf{1}^{\ell k} \in M_N^{\otimes 2} = M_{N^2} \tag{2}$$

(ordinary matrix function).

Consider the equation

$$R(\lambda_1 - \lambda_2) \cdot {}^1L_n(\lambda_1)\, {}^2L_n(\lambda_2) = {}^2L_n(\lambda_2)\, {}^1L_n(\lambda_1) R(\lambda_1 - \lambda_2), \tag{3}$$



Uncertain reading. There is a heavy ink mark between R and ${}^1L_n$. It is represented here by a multiplication dot.

Original source details: GELFAND-P001281

$$ {}^1L_n = L_n \otimes \mathbf{1}, \qquad {}^2L_n = \mathbf{1} \otimes L_n $$

The coefficients are quadratic in the letters $E^q_{\ell k}$

We require (3) to hold identically in λ_1, λ_2.

Eq. (3) is called the YBF equations (Yang–Baxter–Faddeev).



Uncertain reading. In ‘Фаддеева’ (Faddeev), one letter has been overwritten.

Original source details: GELFAND-P001281

It is equivalent to the algebraic system (4)–(5)

$$ [E, E^q_{k\ell}] = 0 \tag{4} $$

$$ [E^p_{ij}, E^q_{k\ell}] = \delta_{jk} E E^{p+q}_{i\ell} - \delta_{\ell i} E E^{p+q}_{kj} + \sum_{a+b=p+q-1}^{a<p\le b} \left(E^a_{kj} E^b_{i\ell} - E^b_{kj} E^a_{i\ell} \right) \tag{5} $$

(6) By definition $E^{n+i}_{k\ell} = 0$ for $i \ge 0$

GELFAND-P001282

[source number: 43]

(7) Claim. For $E = 1$ and on dropping the last term of (5), we obtain

$$ U\big(g\ell_N[t]/(t^n)\big). $$

(envelop. algebra)

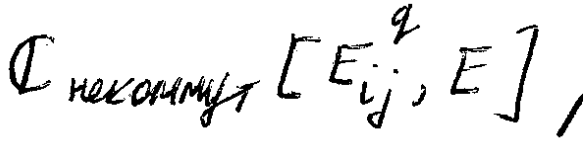


Uncertain reading. Before $g\ell_N$ there is a crossed-out sign, possibly g.

Original source details: GELFAND-P001282

Definition of the Yangian.

$$ Y^n = \mathbb{C}_{\text{noncommut.}}[E^q_{ij}, E] \,/ (4)\text{–}(5) \tag{8} $$

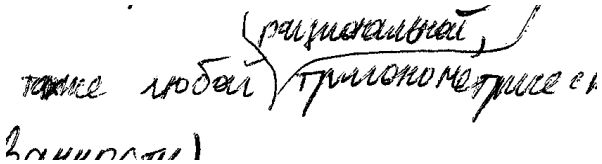


Uncertain reading. Below $\mathbb{C}$ the abbreviation reads ‘некоммут’ (‘noncommutative’). A comma follows E^q_{ij}.

Original source details: GELFAND-P001282

[!] IMG: Idea: commutation relations unlike those in a Lie alg., with $x_1x_2 - x_2x_1$ — a polynomial of deg. 2! Describing quadr. algebras is a proper math. question. (It is interesting when the Poincaré–Birkhoff–Witt theorem holds).

Ch. R-matrix (2) can also be taken to be any [Inserted above the line: rational,] trigonometric and elliptic one (with a compatibility rel.)



Uncertain reading. 'Rational,' is inserted above the line before 'trigonometric'.

Original source details: GELFAND-P001282

[!] Problem (IMG) Classify so that the nth degree has the requisite number of pieces.

Associativity is also optional

(Jackiw: associat. is a cocycle of deg. 3)

Ch.

(9) Claim. For $E = 1$ we have $Y^1 = U(g\ell_N)$

(univ. envelop. algebra)

(10) Zamolodchikov's theorem For rational R the P.B.W. theorem holds if and only if the relation holds $R\,R\,R = R\,R\,R$

Kogan Yan Strings and gauge fields.

Gauge symmetry $\overset{?}{\longleftarrow}$ string

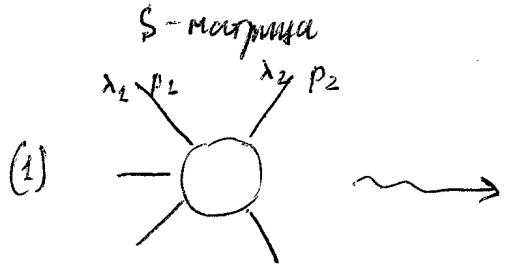
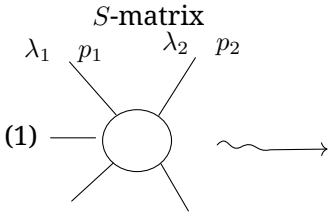


Diagram and typed description.



Original source details: GELFAND-P001282

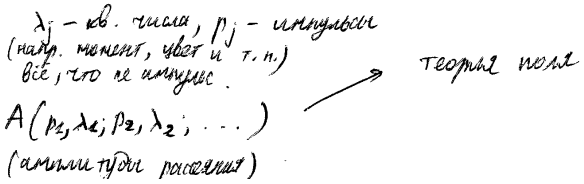
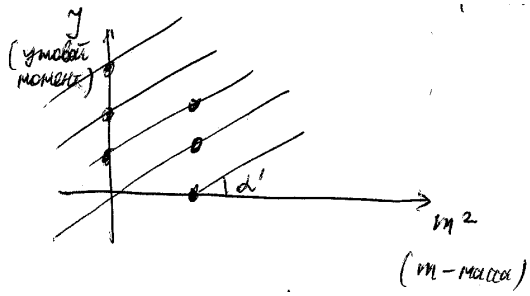


Diagram and typed description.

λ_j — quantum nos., p_j — momenta
(e.g. angular momentum, color, etc.)
everything other than momentum.

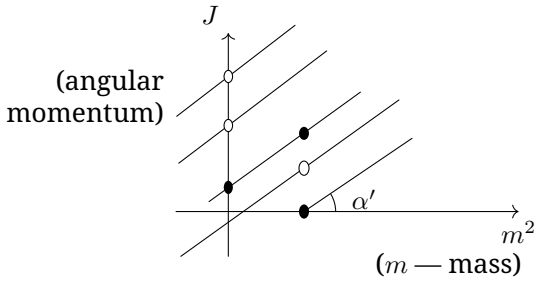
$A(p_1, \lambda_1; p_2, \lambda_2; \ldots)$ → field theory
(scattering amplitudes)

Original source details: GELFAND-P001282



Original source details: GELFAND-P001282

Diagram and typed description.



The six handwritten marks are shown as circles. The amount of fill is uncertain in some marks.

Ang. momentum is always quantized.

(α' — angle —
— string tens.)

[Crossed out]



Original source details: GELFAND-P001282

Uncertain reading. The crossed-out word appears to begin with ‘Угл.’ (‘ang.’).

Limit $\alpha' \to 0 \Rightarrow$ Yang–Mills or Einstein

GELFAND-P001283

Action for the string

$$S = \frac{1}{2\pi\alpha'} \int g^{ab} \sqrt{g}\, \partial_a x^\mu \partial_b x^\mu \, d^2\xi \tag{2}$$

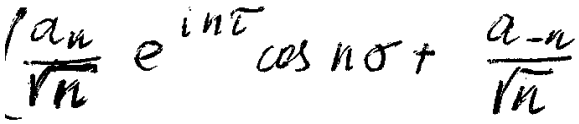
When passing to

$$S = \frac{1}{2\pi\alpha'} \int \partial_a x^\mu \partial_a x^\mu \, d^2\xi \tag{3}$$

the ghost problem arises from the wrong sign for x^0.

[Inserted between the lines: expand in oscillators]

$$x^\mu(\sigma,\tau) = x^\mu + p^\mu \tau + i \sum_{n=1}^{\infty} \left(\frac{a_n}{\sqrt{n}} e^{in\tau} \cos n\sigma + \frac{a_{-n}}{\sqrt{n}} e^{-in\tau} \cos n\sigma \right) \tag{4}$$

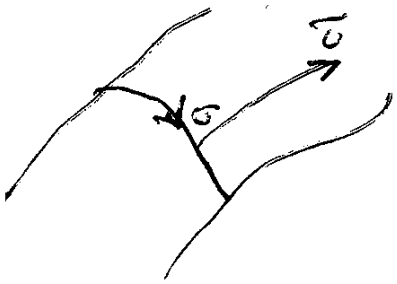


Original source details: GELFAND-P001283

Uncertain reading. Both denominators in (4) read $\sqrt{n}$.

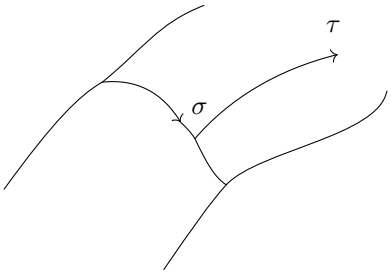
(τ — time)

σ — parameter along the string



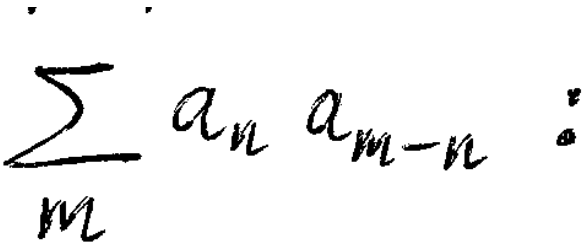
Original source details: GELFAND-P001283

Diagram and typed description.



Virasoro operator

$$L_n =: \sum_m a_n a_{m-n} : \tag{5}$$



Original source details: GELFAND-P001283

Uncertain reading. In (5) the handwritten factors read $a_n a_{m-n}$, with summation over m.

a_k for $k > 0$ — annihilation op.

a_k for $k < 0$ —"— creation

(6) States

$$\underset{\text{(vacuum)}}{|0\rangle} \mapsto \text{tachyon}$$

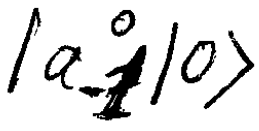
$$a^\mu_{-1}|0\rangle \quad \text{— massless vector particle}$$

$$a^\mu_{-1}a^\nu_{-1}|0\rangle, \qquad a^\mu_{-2}|0\rangle \qquad \text{etc.}$$

In general, it is easy to check that

$$M^2 = -1 + \sum_{n=1}^{\infty} : a^\mu_{-n} a^\mu_n : \tag{7}$$

Many states in (6) are extraneous. For example, $|a^0_{-1}|0\rangle$ has the wrong norm

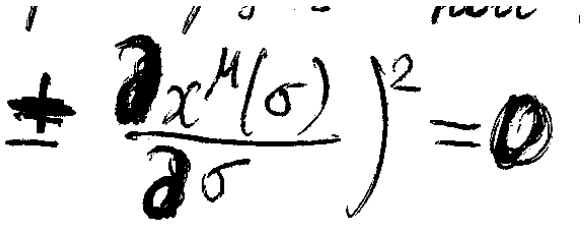


Original source details: GELFAND-P001283

Uncertain reading. The subscript -1 on a is retraced. A separate vertical stroke is visible before a.

[Inserted between the lines: classically there is]
Reparametrization invariance cond.

$$\left(p^{\mu}(\sigma) \pm \frac{\partial x^{\mu}(\sigma)}{\partial \sigma}\right)^2 = 0. \tag{8}$$



Uncertain reading. The $\pm$, the derivative and the zero in (8) are retraced.

Original source details: GELFAND-P001283

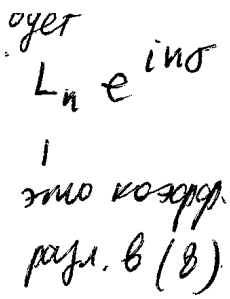
GELFAND-P001284

[source number: 44]

It follows from (8) that

$$\sum_n L_n e^{in\sigma} = 0 \quad \Rightarrow L_n = 0 \tag{9}$$

[Under the first L_n, attached by a short stroke: these are the coeffs. of the expansion in (8).]

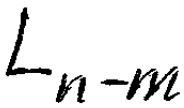


Uncertain reading. A short stroke connects the first L_n to the note below the formula.

Original source details: GELFAND-P001284

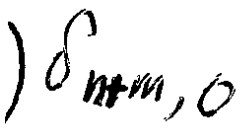
But in quantum theory this is impossible, since there is a central charge.

$$[L_n, L_m] = (n-m)L_{n-m} + \frac{D}{12}(n^3 - n)\delta_{n+m,0} \tag{10}$$



Uncertain reading. The index after $(n-m)$ is written $n-m$.

Original source details: GELFAND-P001284



Uncertain reading. The plus sign in the subscript of δ is retraced.

Original source details: GELFAND-P001284

[Inserted between the lines: note that]

$$[a_n^\mu, a_m^\nu] = n\delta_{n+m,0}\eta^{\mu\nu} \tag{11}$$

The condition on the state

$$L_n|\Psi\rangle = 0 \qquad \text{for all } n. \tag{12}$$

cannot be imposed.

Impose the conditions

$$L_n|\Psi\rangle = 0, \quad n \geq 1, \quad (L_0 - 1)|\Psi\rangle = 0 \tag{13}$$

The class. conditions (9) are to be interpreted through expectation values.



Uncertain reading. The first letter of ‘Классич.’ (‘class.’) is retraced.

Original source details: GELFAND-P001284

$$\langle\Psi|L_n|\Psi\rangle = 0, \tag{14}$$

which follows from (13) for any $n \neq 0$.

(15) String theory — quantum mech. of a large Hilbert space in which conds. (13) are imposed.



Uncertain reading. The abbreviation ‘к-ром’ (‘which’) has been overwritten.

Original source details: GELFAND-P001284

This is the old string theory.

Now string field theory.

(First in ’71–’72)

2 ways

Consider $|0\rangle \rightsquigarrow \varphi(x)$

$a_{-1}^\mu|0\rangle \rightsquigarrow A^\mu(x)$



Uncertain reading. A subscript on $A^\mu(x)$, possibly μ, has been crossed out.

Original source details: GELFAND-P001284

$$a_{-1}^\mu a_{-1}^\nu|0\rangle; \quad a_{-2}^\mu|0\rangle \rightsquigarrow h^{\mu\nu}(x), B^\mu(x)$$

In more accessible terms, consider

$$\left(\varphi(x) + A_\mu(x)a^\mu_{-1} + h_{\mu\nu}(x)a^\mu_{-1}a^\nu_{-1} + B_\mu(x)a^\mu_{-2} + \ldots\right)|0\rangle \tag{16}$$

In the old notation using $x^\mu(\sigma) \mapsto x^\mu, a^\mu_n$
this was a functional of the contour.
(16) — a state (a set of states),
to a point $x(\sigma) \rightsquigarrow$ the state $\Phi(x(\sigma))$. (field on the string)
An analogous expansion can be obtained

$$\Phi[x(\sigma)] = \Phi[x, a^\mu_n]$$

(17) Rule for the transition
Expand in a Taylor series in a^μ_n,
the coeffs. of a^μ_n with $n > 0$ are inessential (nonphysical),
discard them and retain only the operators of creation.
[Crossed out before (18)]



Uncertain reading. The crossed-out fragment may read ‘Из’ (‘From’).

Original source details: GELFAND-P001285

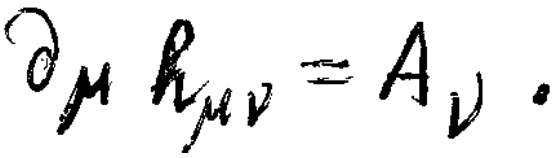
(18) There is reparametriz. invariance under conditions (13).
$\Rightarrow$ the fields are interdependent.
[Inserted between the lines: calculating by formula (5), we obtain]

$$L_1 = a^\mu_1 \frac{\partial}{\partial x^\mu} + \ldots \tag{19}$$

Applying this to (16), we obtain

$$\partial_\mu A_\mu(x) = 0, \tag{20}$$

$$\partial_\mu h_{\mu\nu} = A_\nu. \tag{21}$$



Uncertain reading. The right side of (21) is written A_ν.

Original source details: GELFAND-P001285

(up to coefficients)
This gives the Lorenz gauge cond. (20).
Thus, instead of imposing constraints, consider

$$\delta|\Psi\rangle = \sum_{n=1}^{\infty} b_n L_{-n}|\Phi\rangle, \tag{22}$$

[Below δ, with a short connecting stroke: variation of the state]
b_n — arbitrary parameters.
We will proceed so as to maintain this invariance.

$$\delta A_\mu = \partial_\mu \varepsilon + [\varepsilon, A] \tag{23}$$



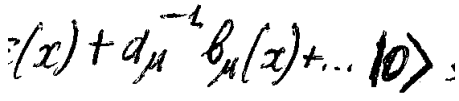
Uncertain reading. In the commutator, A is written without an index, after ε.

Original source details: GELFAND-P001285

GELFAND-P001286

[source number: 45]

$$\begin{aligned}(\delta\varphi(x) + \delta A_\mu(x) a_\mu^{-1} \ldots)|0\rangle = \\ b_1 \frac{\partial}{\partial x^\mu} a_\mu^{-1}(\varepsilon(x) + a_\mu^{-1} b_\mu(x) + \ldots|0\rangle = \\ = (\ldots\ b_1 \partial_\mu \varepsilon(x) a_\mu^{-1} + b_1 \partial_\mu b_\nu a_{-1}^\mu a_{-1}^\nu\ \ldots|0\rangle\end{aligned} \tag{24}$$





Uncertain reading. No closing round parenthesis is visible before $|0\rangle$ in either right-hand expression.

Original source details: GELFAND-P001286

and in the end we obtain

$$\delta A_\mu = \partial_\mu \varepsilon(x) \tag{25}$$

which gives a genuine gauge field.
Thus, [crossed out]

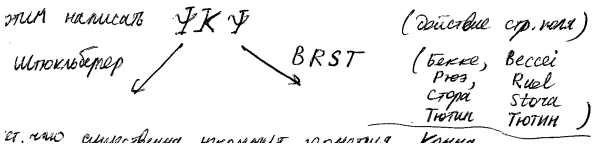


Undeciphered passage. The crossed-out mark after ‘Thus,’ is undeciphered.

Original source details: GELFAND-P001286

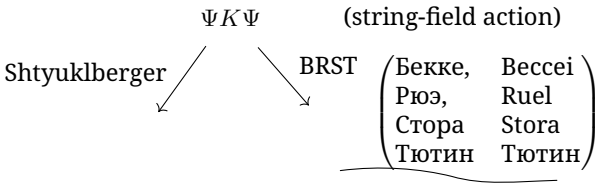
2-dimensional reparam. invariance turns into gauge symmetry.

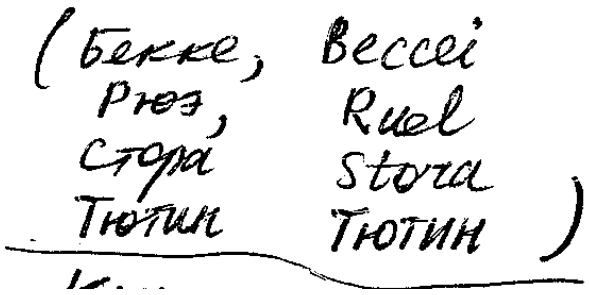
Next, we want to write



Original source details: GELFAND-P001286

Diagram and typed description.





Original source details: GELFAND-P001286

Uncertain reading. The right column reads ‘Beccei’, ‘Ruel’, ‘Stora’, ‘Тютин’.



Original source details: GELFAND-P001286

Uncertain reading. The label by the left arrow reads ‘Штюкльбергер’ (‘Shtyuklberger’).

Witten writes that Connes’s noncommut. geometry is essential.

(nonabelianization, etc.)

GELFAND-P001287

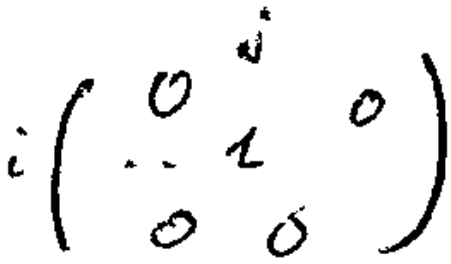
[source number: 61]

28.4.86 Cherednik. Yangian.



Original source details: GELFAND-P001287

Uncertain reading. The manuscript date is ‘28.4.86’. Catalogue record 676 gives 28 April 1987.



Original source details: GELFAND-P001287

Diagram and typed description.

$M_N \ni \mathbf{1}$, $\mathbf{1}^{ij}$ — std. matrix

$$i \begin{pmatrix} 0 & & 0 \\ \cdot\cdot & 1 & \\ 0 & 0 & \end{pmatrix}^{j} \quad N \times N$$

$$L = \mathbf{1} + \sum_{\substack{q=1 \\ 1\le i,j\le N}}^{n} \lambda^{-q} \mathbf{1}^{ij} E_{ji}^{q-1}, \tag{1}$$

E_{ji}^{q-1} [crossed out] — formal variables



Undeciphered passage. The word before ‘formal variables’ is obscured by crossing-out strokes.

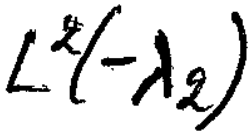
Original source details: GELFAND-P001287

n, N are fixed

$$R = \mathbf{1} \otimes \mathbf{1} + \frac{1}{\lambda} \sum \mathbf{1}^{ij} \otimes \mathbf{1}^{ji} \in M_N^{\otimes 2} \tag{2}$$

$$\begin{aligned} \text{(YBF)} \quad & R(\lambda_1 - \lambda_2) L^1(\lambda_1) L^2(-\lambda_2) = L^2(\lambda_2) L^1(\lambda_1) R(\lambda_1 - \lambda_2), \\ & L^1 = L \otimes \mathbf{1}, \qquad L^2 = \mathbf{1} \otimes L \end{aligned} \tag{3}$$

(Yang–Baxter–Faddeev)



Uncertain reading. On the left there is a minus in $L^2(-\lambda_2)$.

Original source details: GELFAND-P001287

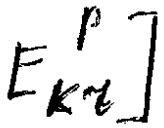


Uncertain reading. On the right the factor reads $L^2(\lambda_2)$.

Original source details: GELFAND-P001287

$$\Updownarrow$$

$$\begin{aligned} [E_{ij}^q, E_{k\tau}^p] = {} & \delta_{jk} E_{i\tau}^{p+q} - \delta_{\tau i} E_{kj}^{p+q} \\ & + \sum_{a<p\le b}^{a+b=p+q-1} (E_{kj}^a \cdot E_{i\tau}^b - E_{kj}^b \cdot E_{i\tau}^a) \end{aligned} \tag{*}$$



Uncertain reading. The fourth matrix index is read as τ.

Original source details: GELFAND-P001287

This is a quantum algebra generalizing Lie algebra. If we insert the parameter $\mathbf{1} \rightsquigarrow i\mathbf{1}$, then in the limit we obtain the matrix Lie alg..



Uncertain reading. The sign before **1**, read as i, has a dot above it.

Original source details: GELFAND-P001287

Definition 1. Yangian of level n

$$Y_N^n = \mathbb{C}'[E_{ij}^q]/(*) \tag{4}$$

[Under $\mathbb{C}'$, with a short connecting stroke: ring of noncommut. polynomials]

$$Y_N^1 = U(g\ell_N), \qquad Y_N^1 \ni E_{ij}^0 \longrightarrow e_{ij} \in g\ell_N \tag{5}$$

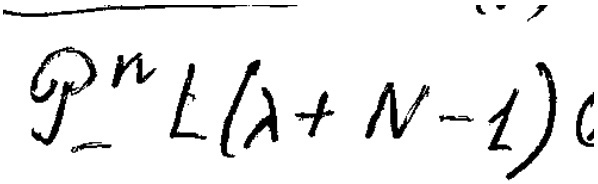
If the indices are outside the range of def., then $E_{\cdot\cdot}^{\cdot} = 0$, i.e.

$$E_{ij}^a = 0 \quad \text{for} \quad \begin{matrix} a < 0, \\ a > n-1 \end{matrix}$$

Definition 2. $H(\lambda)$ is defined from the relations

$$\mathcal{P}_-^n L(\lambda + N - 1) \otimes \ldots \otimes L(\lambda)\mathcal{P}_-^n \overset{\text{def}}{=} H(\lambda)\mathcal{P}_-^n \tag{6}$$

[Under $H(\lambda)$, bracketed: this is the determinant]



Uncertain reading. The antisymmetrizer bears n, and the first shift is $N-1$.

Original source details: GELFAND-P001287

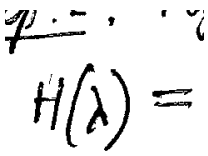
$\mathcal{P}_-^n$ — the antisymmetrizer $\in (M_N)^{\otimes n}$, acting in $(\mathbb{C}^N)^{\otimes n}$

$$\mathcal{P}_-^2 = \frac{1}{2}(\mathbf{1} \otimes \mathbf{1} - P)$$

$H(\lambda) \in Y_N^n$; on the right of (6) we obtain a scalar matr.

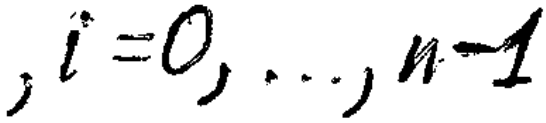
Thm. 1. Let

$$H(\lambda) = \sum_{q=1}^{n} (\lambda + i)^{-q} H_i^q + O(1) \quad \text{(in a nbhd of } \lambda = -i), \quad i = 0, \ldots, n-1 \tag{7}$$



Uncertain reading. The equals sign before the sum in (7) has two strokes.

Original source details: GELFAND-P001287



Uncertain reading. The upper end of the range of i is written $n-1$.

Original source details: GELFAND-P001287

Then the center of Y_N^n is given by the formula

$$c(Y_N^n) = \mathbb{C}[H_i^q] \tag{8}$$

GELFAND-P001288

In the absence of shifts [crossed out] (6) would be a pure def. of det.



Undeciphered passage. A word before '(6)' is crossed out. Its exact reading is uncertain.

Original source details: GELFAND-P001288

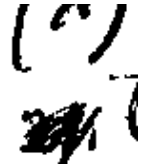
$$H(\lambda) \approx \det L.$$



Uncertain reading. The two wavy strokes between $H(\lambda)$ and $\det L$ are read as $\approx$.

Original source details: GELFAND-P001288

Rem. If [crossed out] U — a scalar matrix (if the letters in U commute), then



Undeciphered passage. A short crossed-out sign follows 'Если' ('If').

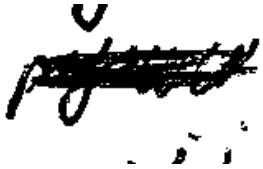
Original source details: GELFAND-P001288

$$\mathcal{P}_-^N \left(\mathbf{1} + \frac{1}{\lambda} U\right)^{\otimes N} \mathcal{P}_-^N = \det\left(\mathbf{1} + \frac{1}{\lambda} U\right) \mathcal{P}_-^N \tag{9}$$

Calculation of $H(\lambda)$ on the univ. envelop. algebra. Now

$$L = \mathbf{1} + \frac{1}{\lambda} \sum \mathbf{1}^{ij} e_{ji}, \qquad e_{ij} \in U(g\ell_N) \tag{10}$$

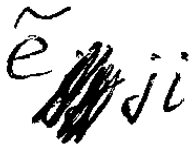
Representation $e_{ij} \longrightarrow \widetilde{e}_{ij} \in M_d$, satisf. [crossed out] the usual relations



Undeciphered passage. The word after 'удовл.' ('satisf.') is crossed out. Its reading is uncertain.

Original source details: GELFAND-P001288

$$V : g\ell_N \longleftrightarrow L_V = \mathbf{1} + \frac{1}{\lambda}\sum \mathbf{1}^{ij}\widetilde{e}_{ji} \in M_N(M_d) \tag{11}$$



Uncertain reading. The first subscript on $\widetilde{e}$ is crossed out, possibly ij. To its right is ji.

Original source details: GELFAND-P001288

V — a finite-dimensional rep. of $g\ell_N$ ($g\ell_N$-module).

From the $g\ell_N$-module V we construct the matrix func. $L_V(\lambda)$

$$\widetilde{e}_{ij} \in \operatorname{End} V = M_d$$

Example.

$$V = \mathbb{C}^N \longrightarrow L_{id} = \mathbf{1} + \frac{1}{\lambda}\sum \mathbf{1}^{ij} \otimes \mathbf{1}^{ji} \in M_N(M_N)$$

and there is an identification of tensor prod. and matrices

[crossed out]

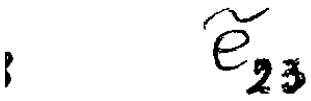


Uncertain reading. The crossed-out word appears to be ‘Вопросы’ (‘Questions’).

Original source details: GELFAND-P001288

$$\begin{pmatrix} I + \frac{1}{\lambda}\widetilde{e}_{11} & \frac{1}{\lambda}\widetilde{e}_{21} & \dots \\ \frac{1}{\lambda}\widetilde{e}_{12} & I + \frac{1}{\lambda}\widetilde{e}_{22} & \dots \\ \frac{1}{\lambda}\widetilde{e}_{13} & \widetilde{e}_{23} & \dots \\ \cdot\ \cdot & \cdot & \cdot\ \cdot \end{pmatrix}$$

— the matrix on the right in (11)



Uncertain reading. No factor $1/\lambda$ is written before $\widetilde{e}_{23}$.

Original source details: GELFAND-P001288

(12) Problem. Compute L_V in terms of L_{id} for any V.

Let an irred. rep. assoc. with a Young diagram be given

$$\mu = (m_1 \geq m_2 \geq \ldots \geq m_\tau > 0), \quad \tau \leq N, \quad \sum m_i = n \tag{13}$$

$$V_\mu = \mathcal{P}_\mu(\mathbb{C}^N)^{\otimes n} \tag{14}$$

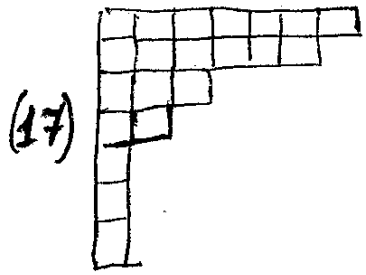
$\mathcal{P}_\mu$ — the Young projector (it is written out explicitly)

$$L_\mu = \mathbf{1} + \frac{1}{\lambda}\sum \mathbf{1}^{ij}\mathcal{P}_\mu e_{ji}\mathcal{P}_\mu \tag{15}$$

$$L_{V_\mu} = \widetilde{\mathcal{P}}_\mu\{L_{id}(\lambda - u_1)\cdot L_{id}(\lambda - u_2)\cdot \ldots \cdot L_{id}(\lambda - u_n)\}\widetilde{\mathcal{P}}_\mu \tag{16}$$

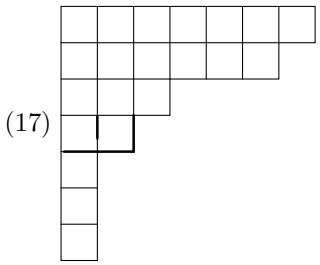
GELFAND-P001289

[source number: 62]



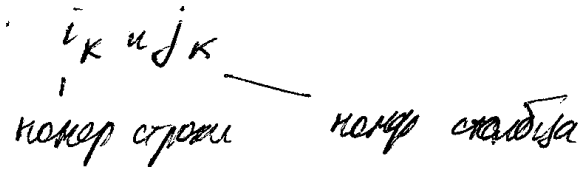
Original source details: GELFAND-P001289

Diagram and typed description.



The numbers from 1 to n are written into the diagram — we obtain — a std. tableau.

Each k acquires



Original source details: GELFAND-P001289

Diagram and typed description.

i_k and j_k

row number column number

$$u_k = i_k - j_k, \qquad u_1 = 0$$

Def. of the bold dot in (16).

The matrices have different coefficient rings. In multiplying them we take, in the matrix entries,

$$A \bullet B = A \otimes B$$

$\mathcal{P}_\mu$ acts only on the matrix coefficients ($\mathcal{P}_\mu$ — a scalar matrix)

$$\mathcal{P}_\mu = \mathcal{P}_\mu \mathbf{1}$$



Original source details: GELFAND-P001289

Uncertain reading. A superscript mark on the first $\mathcal{P}_\mu$ is crossed out.

def.

$$\mathcal{P}_\mu \in M_N^{\otimes n}$$

$\mathcal{P}_\mu$ — symmetrizer along the rows and antisymm. along the columns

On $M_N^{\otimes n}$ acts the symm. group S_n ord. n



Original source details: GELFAND-P001289

Uncertain reading. After S_n the manuscript reads ‘пор. n’ (‘ord. n’).

GELFAND-P001290

5/$\overline{\text{V}}$–86 G. Olshanskii. Infinite-dimensional reps. and the Yangian.

(Gelfand sem.)



Original source details: GELFAND-P001290

Uncertain reading. The manuscript date is ‘5/V–86’. Catalogue record 677 gives 5 May 1987.

$o(n)$ — complex skew-symm. matrices $n \times n$

$$o(\infty) = \bigcup_{n=1}^{\infty} o(n),$$

ρ — tautological rep. of $o(\infty)$, acting on $\ell_2 = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix} = H$

$$\rho^{\otimes k} = \underbrace{\rho \otimes \cdots \otimes \rho}_{k \text{ times}},$$



Original source details: GELFAND-P001290

Undeciphered passage. A mark to the right of the tensor power is crossed out.

$$A = \Big\{ X = (X_n),\ X_n \in U(o(n))$$

[Below U] univ. envelop. alg.



Original source details: GELFAND-P001290

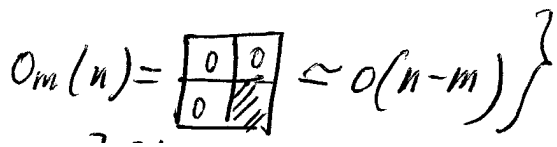
Undeciphered passage. A crossed-out mark stands by the explanation of U.

props.

1) convergence $\exists\, s\text{-}\lim\limits_{n\to\infty} \rho^{\otimes k}(X_n)$ in $H^{\otimes k}$, $\forall k$ (the limit is denoted $\rho^{\otimes k}(X)$)

2) const. $\exists M = M(X):\ \deg X_n \le M$

3) finite index $\exists m = m(X):\ X_n \in U(o(n))^{O_m(n)}$



Original source details: GELFAND-P001290

Diagram and typed description.

$O_m(n) = \begin{array}{|c|c|}\hline 0 & 0 \\ \hline 0 & \\ \hline \end{array} \simeq o(n-m) \Big\}$



Undeciphered passage. A mark in the left margin before ‘Theorem’ is crossed out.

Original source details: GELFAND-P001290

Theorem.

$$A = \mathbb{C}[x_1, x_2, \ldots] \otimes Y,$$

where Y — an alg. with generators $t_{ij}^{(M)}$, $M, i, j = 1, 2, \ldots$

$$(1) \quad [t_{ij}^{(M+1)}, t_{k\ell}^{(N)}] - [t_{ij}^{(M)}, t_{k\ell}^{(N+1)}] = t_{kj}^{(M)} t_{i\ell}^{(N)} - t_{kj}^{(N)} t_{i\ell}^{(M)}$$

Remark Y — the Yangian [inserted above with a line to this word] (for $g\ell(m)$).



Uncertain reading. The argument of $g\ell$ in the insertion is m.

Original source details: GELFAND-P001290

In fact this is an operator realization of the Yangian.



Uncertain reading. The abbreviation before ‘реализ.’ reads ‘операт.’ (‘operator’).

Original source details: GELFAND-P001290

4th prop. Sequences with identical limits in all representations are identified.

$$X \sim Y, \quad \text{if} \quad \rho^{\otimes k}(X) = \rho^{\otimes k}(Y)$$

$m = 0 \quad \Rightarrow$ 3) picks out the center

$m = 1 \quad \Rightarrow$ 3) [crossed out] matr. commuting with



Undeciphered passage. A short mark following ‘3)’ is crossed out.

Original source details: GELFAND-P001290



Diagram and typed description.

$\begin{matrix} 0 & 0 \\ 0 & \end{matrix}$

— a problem in lin. algebra.

Original source details: GELFAND-P001290

$Y = \bigcup Y_m$ We may fix m.

m is fixed $\Rightarrow$ we obtain

$$A_m = \mathbb{C}[x_1, x_2, \ldots] \otimes Y_m$$

$$t_{ij}^{(M)} \qquad (i, j \leq m;\ M = 1, 2, \ldots)$$

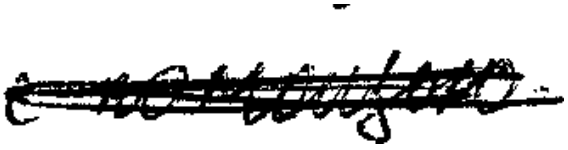
GELFAND-P001291

[source number: 63]

For $sp(\infty)$ we obtain the same A

For $u(\infty)$ —"— $A \otimes A$.

Thus the same A arises throughout, and there is an explanation [crossed out] from representation theory.

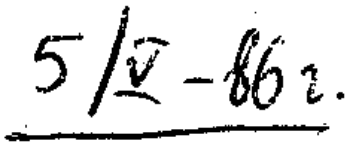


Original source details: GELFAND-P001291

Uncertain reading. The crossed-out writing may read ‘с помощью’ (‘by means of’).

Proposal I.M.G. to consider first the cases $m = 0$, $m = 1$

5/V–86 Cherednik I.



Original source details: GELFAND-P001291

Uncertain reading. The manuscript date is ‘5/V–86’. Catalogue record 678 gives 5 May 1987.

$$\mathbf{1},\ \mathbf{1}^{ij} \in M_N, \qquad e_{ij} \in U(g\ell_N); \qquad M_N \text{ — matrices } N \times N;$$

$$L(\lambda) = \mathbf{1} + \frac{1}{\lambda} \sum_{i,j} e_{ij} \mathbf{1}^{ji} \in M_N(U(g\ell_N));$$

$$H(\lambda) = \det_Q L(\lambda);$$

(see def. below) (Q from the word quantum)

$$\rho : g\ell_N \longrightarrow \operatorname{End} V \simeq M_d \qquad \text{(rep.)};$$

$$L_V = \mathbf{1} + \frac{1}{\lambda} \sum_{i,j} \rho(e_{ij}) \mathbf{1}^{ji} \in M_N(\operatorname{End} V).$$

Let V — be a rep. with [crossed out] highest weight $\mu = (m_1, \ldots, m_\tau, 0, \ldots, 0)$



Original source details: GELFAND-P001291

Undeciphered passage. A short mark before ‘старшим’ (‘highest’) is crossed out.

$$m_1 \geq m_2 \geq \cdots \geq 0, \qquad \tau \leq N, \qquad m_j \in \mathbb{Z}_+$$

$$V = V_\mu$$

For the group ∃! a vector, invariant under



Uncertain reading. The symbols ∃! stand before ‘vector’.

Original source details: GELFAND-P001291

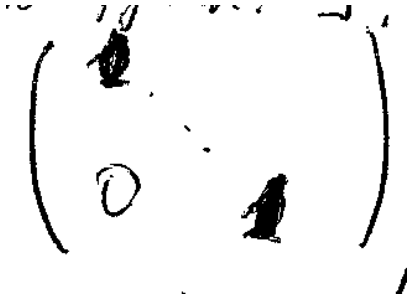


Diagram and typed description.

$$\begin{pmatrix} 1 & & \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

Original source details: GELFAND-P001291

Then for

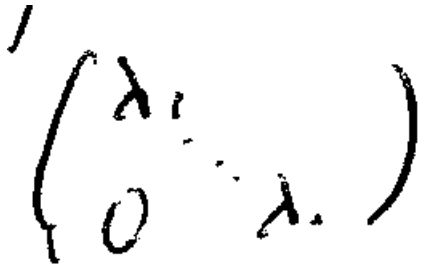


Diagram and typed description.

$$\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ 0 & & \lambda_\cdot \end{pmatrix}$$

A small dot follows the lower λ. No index letter is distinct.

Original source details: GELFAND-P001291

it is an eigenvector with [crossed out] eigenvalue $\lambda_1^{m_1}\lambda_2^{m_2}\cdots\lambda_\tau^{m_\tau}$

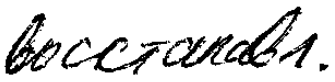


Undeciphered passage. The mark before ‘соб. знач.’ (‘eigenvalue’) is crossed out.

Original source details: GELFAND-P001291

and necessarily $m_1 \geq m_2 \geq \cdots \geq m_\tau$.

From it, uniquely reconstr. How?



Uncertain reading. The abbreviation reads ‘восстанавл.’ (‘reconstr.’).

Original source details: GELFAND-P001291

Construction by means of

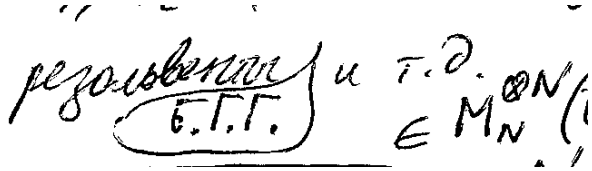


Diagram and typed description.

resolvents etc.

B.G.G.

Original source details: GELFAND-P001291

Def. of $H(\lambda)$

$$(1)\quad H(\lambda)\mathcal{P}_-^N = \mathcal{P}_-^N\big(L(\lambda+N-1)\otimes\cdots\otimes L(\lambda)\big)\mathcal{P}_-^N,$$
$$H(\lambda)\in U(g\ell_N)$$

[Above the tensor product, a bracket labelled $\in M_N^{\otimes N}(U(g\ell_N))$.]

(2) Thm.

$$H(\lambda)\big|_{V_\mu} \overset{\text{def}}{=} \det_Q L_{V_\mu} = \prod_{i=1}^{\tau}\frac{\lambda+N+m_i-i}{\lambda+N-i}$$

(in particular, it is asserted to be a scalar func.)

$\mathcal{P}_-^N$ — the antisymmetrizer

$$\mathcal{P}_-^N(x_1\otimes\cdots\otimes x_N) = \frac{1}{N!}\sum_{\sigma\in S_N}(-1)^{\epsilon(\sigma)}(x_{\sigma(1)}\otimes\cdots\otimes x_{\sigma(N)}),$$
$$x_j\in\mathbb{C}^N$$

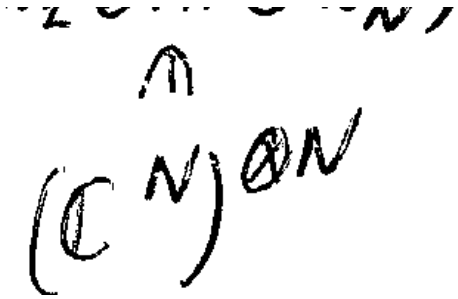


Diagram and typed description.

$$x_1\otimes\cdots\otimes x_N$$
$$\big(\mathbb{C}^N\big)^{\overset{\wedge}{\otimes} N}$$

A curved mark with three strokes stands below the argument. It may be a rotated $\in$.

Original source details: GELFAND-P001291

GELFAND-P001292

$$\mathcal{P}_-^N\in M_N^{\otimes N} = \mathrm{End}\big((\mathbb{C}^N)^{\otimes N}\big)$$

The image of $\mathcal{P}_-^N$ is a one-dimensional subspace.

On the right in (1) is the restriction of $L(\lambda+N-1)\otimes\cdots\otimes L(\lambda)$ to the 1-dimensional subspace.



Uncertain reading. The earlier lettering beneath the overwritten L in the last factor is illegible.

Original source details: GELFAND-P001292

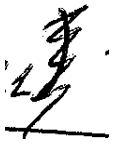
$$H\mathcal{P}_-^2 = \mathcal{P}_-^2(A \otimes B)\mathcal{P}_-^2$$

$$\underbrace{\frac{1}{4}\left(a_{11}b_{22} + a_{22}b_{11} - a_{12}b_{21} - a_{21}b_{12}\right)}_{H \text{ in the case of arbitrary matrices}}$$



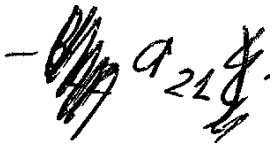
Original source details: GELFAND-P001292

Uncertain reading. Before $\frac{1}{4}$ is a cancelled sign, possibly $=$. Between the fraction and the parenthesis is an obscured stroke.



Original source details: GELFAND-P001292

Uncertain reading. After $a_{22}b_{11}$ a mark is cancelled, possibly $\frac{1}{2}$.



Original source details: GELFAND-P001292

Uncertain reading. Before a_{21} is a cancelled entry, possibly b_{12}. Between a_{21} and b_{12} a mark is cancelled, possibly $\frac{1}{2}$.

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}.$$

$$a_{11} = 1 + \frac{1}{\lambda+1}e_{11}, \quad a_{12} = \frac{1}{\lambda+1}e_{21},$$
$$a_{22} = 1 + \frac{1}{\lambda+1}e_{22}, \quad a_{21} = \frac{1}{\lambda+1}e_{12}.$$



Original source details: GELFAND-P001292

Uncertain reading. The earlier letter under the correction in a_{21} is illegible.

$$b_{11} = 1 + \frac{1}{\lambda} e_{11}, \quad b_{12} = \frac{1}{\lambda} e_{21},$$
$$b_{22} = 1 + \frac{1}{\lambda} e_{22}, \quad b_{21} = \frac{1}{\lambda} e_{12}.$$

$$\begin{aligned} H(\lambda) = & \left(1 + \frac{1}{\lambda+1} e_{11}\right)\left(1 + \frac{1}{\lambda} e_{22}\right) \\ & + \left(1 + \frac{1}{\lambda+1} e_{22}\right)\left(1 + \frac{1}{\lambda} e_{11}\right) - \\ & - \left(\frac{1}{\lambda+1} e_{12}\right)\left(\frac{1}{\lambda} e_{21}\right) - \left(\frac{1}{\lambda+1} e_{21}\right)\left(\frac{1}{\lambda} e_{12}\right). \end{aligned}$$



Uncertain reading. The writing in $H(\lambda)$ before the equals sign is overwritten and the earlier strokes are illegible.

Original source details: GELFAND-P001292



Uncertain reading. Before the minus at the line end, an indexed e is cancelled, possibly e_{11} or e_{22}.

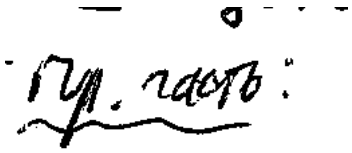
Original source details: GELFAND-P001292

Claim. $H(\lambda)$ is a multiple of 1 on every irreducible rep. ($\in$ the center).



Uncertain reading. An illegible entry is cancelled before the heading.

Original source details: GELFAND-P001292

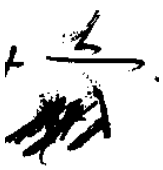


Uncertain reading. Пр. part (the abbreviation is uncertain).

Original source details: GELFAND-P001292

(G. Olshanskii)

$$\begin{aligned} H(\lambda) = & \sum \operatorname{sgn} \begin{pmatrix} i_1 & \cdots & i_N \\ j_1 & \cdots & j_N \end{pmatrix} \left(\delta_{i_1 j_1} + \frac{1}{\lambda + N - 1} e_{i_1 j_1}\right) \\ & \left(\delta_{i_2 j_2} + \frac{1}{\lambda + N - 2} e_{i_2 j_2}\right) \cdots \left(\delta_{i_N j_N} + \frac{1}{\lambda} e_{i_N j_N}\right) \end{aligned}$$



Uncertain reading. Illegible marks are cancelled before λ in the last denominator.

Original source details: GELFAND-P001292

If the letters e_{ij} commute (introduce h) $\Rightarrow$ we obtain the charact. polynomial.

If noncommut. this is the correct analogue of the charact. polynomial.

The following commute: $\sum e_{ii}$, $\sum e_{ij}e_{ji}$, etc.

GELFAND-P001293

1/IX - 86 Gelfand Sem.

At Princeton Witten gave a course on strings, and he had 2 listeners, Deligne and Kazhdan. ['strings' is underlined with a wavy line.]

I.M.G. and Serganova $\rightarrow$ matroids



Uncertain reading. Beneath the right-pointing arrow is an unclear sign, possibly a colon or a cancelled stroke.

Original source details: GELFAND-P001293

Beilinson and Ginzburg: quadr. algebras

Cherednik: Gelfand–Tsetlin basis

[The names in the three lines above are underlined with a wavy line. At left is a broken vertical line.]

I.M. Gelfand (and Serganova).

General definition of a matroid. [Underlined with a wavy line.]

Homolog. algebra is a branch of combinatorics

In Aigner's book 'Combinat. Theory' a third is devoted to matroids. ['Aigner' is underlined with a wavy line.]

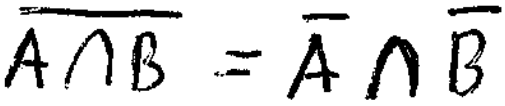
Def. 1. Matroid.

A finite set and a notion of 'closure' on it:

$$A \rightsquigarrow \bar{A} \supset A$$

such that

a) $\overline{A \cap B} = \bar{A} \cap \bar{B}$ (but this is not required for $A \cup B$)



Uncertain reading. The manuscript has an equals sign between $\overline{A \cap B}$ and $\bar{A} \cap \bar{B}$.

Original source details: GELFAND-P001293

b) [crossed out] $\bar{\bar{A}} = \bar{A}$.



Uncertain reading. The crossed-out formula may read $a \in A$ or $a \in \bar{A}$.

Original source details: GELFAND-P001293

[Inserted at right, above the crossed-out line.]

$$\text{c)} \quad a \notin \bar{A}, \quad a \in \overline{A + \{b\}} \quad \Rightarrow \quad b \in \overline{A + \{a\}}.$$

[[crossed out] c) $a \in \bar{A},\ b \notin \bar{A}\ \{a \in A,\ b \in \bar{A},\ b \neq a\} \Rightarrow a \in \overline{A \setminus \{b\}}$.]



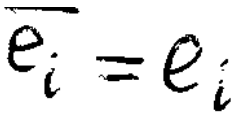
Uncertain reading. The start of the crossed-out line reads $a \in \bar{A},\ b \notin \bar{A}$, with crossing-out strokes over the membership signs.

Original source details: GELFAND-P001293

Example. In $\mathbb{R}^n$ consider k vectors $e_1, \ldots, e_k$.

$$A \subset \{1, 2, \ldots, k\};$$
$$\bar{A} \ni i \ \Leftrightarrow\ e_i \in \text{lin. span of } \{e_j,\ j \in A\}$$

$\bar{e}_i = e_i$; but $\overline{e_i \cup e_j} \neq \bar{e}_i \cup \bar{e}_j$ (in general).



Uncertain reading. The manuscript reads $\bar{e}_i = e_i$.

Original source details: GELFAND-P001293

Nondeg. matroid (or combinat. geometry) if $\bar{\varnothing} = \varnothing$. [Underlined with a wavy line.]

Where does a matroid arise?

Thm. (Gelf., Serg., Goresky, MacPherson). The Grassmann set is divided into strata (under the action of the Cartan subgroup), and there are also Schubert cells.



Uncertain reading. The surname appears to be written ‘Горецки’ (Goresky).

Original source details: GELFAND-P001293

In $\mathbb{R}^n$ diag. matrices act $\Rightarrow$ action on the Grassm.

The orbits are grouped into similar ones. These are the strata.

GELFAND-P001294

[source number: 71]

Each stratum in $G_{n,k}$ is uniquely specified by a matroid.

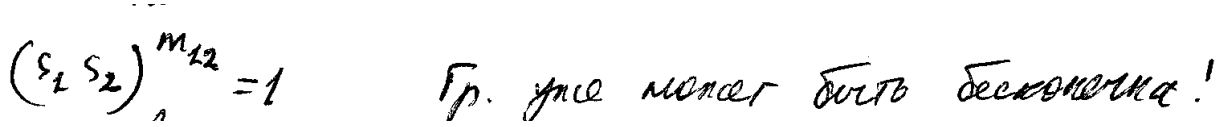
General def. of a matroid

Let a Coxeter grp. be given, gen. by $s_1, \dots, s_n$ (i.e., there are rel. $s_i^2 = 1$, $(s_i s_j)^{m_{ij}} = 1$, m_{ij} a given integer matrix)

Examples.

A. $1, s_1$ — 2 elements

B. $1, s_1, s_2$ — $(s_1 s_2)^{m_{12}} = 1$ The grp. can already be infinite!



Original source details: GELFAND-P001294

Uncertain reading. Both the relation with exponent m_{12} and the words 'The grp. can already be infinite!' are written.

Question. When? (for first-year students)

In general. When is it finite? Already difficult.



Original source details: GELFAND-P001294

Uncertain reading. Above 'In general' is a wavy stroke whose function is unclear.

Example. [wavy underlining] the permutation grp. S_n on n elements;



Original source details: GELFAND-P001294

Uncertain reading. 'Подстановок' (permutations) has thickened strokes in the middle.

s_j — [crossed out] transpos. of 2 elements (adjacent). (j and $j+1$)



Original source details: GELFAND-P001294

Uncertain reading. The crossed-out word may read 'это' ('is').

$$(s_i s_{i+1})^3 = 1, \qquad (s_i s_k)^2 = 1 \quad \text{for nonadjacent } i \text{ and } k.$$

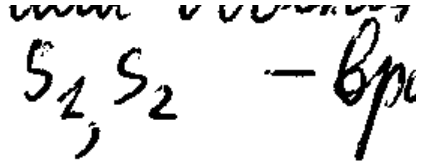
H^3: s_1, s_2, s_3:

$$(s_1 s_2)^5 = 1, \qquad (s_2 s_3)^3 = 1, \qquad (s_1 s_3)^2 = 1 \qquad \text{(120 elements)}$$

(symmetry grp. of the icosahedron!)

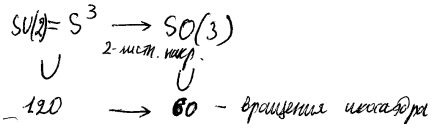
[Between the lines] or of the dodecahedron.

s_1, s_2 — rot. about a vertex



Original source details: GELFAND-P001294

Uncertain reading. There is a comma between s_1 and s_2.



Original source details: GELFAND-P001294

Diagram and typed description.

$$\begin{array}{ccc} SU(2) = S^3 & \xrightarrow[\text{2-sheeted cover}]{} & SO(3) \\ \cup & & \cup \\ 120 & \longrightarrow & 60 \end{array}$$

— rotations of the icosahedron

(See ch. 1 of the book by Gelf. — Minlos — Shapiro)



Original source details: GELFAND-P001294

Diagram and typed description.



The upper stroke starts below 120 and curves around the reference line at left. The lower stroke ends in an arrow before '120 points'. A join is indistinct.

120 points in $S^3 = SU(2)$

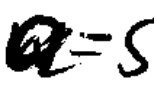
Convex hull — a regular polyh. with 600 faces, tetrahedra

Its symmetry grp. H^4. Its order is 120^2.

Flag matroid.

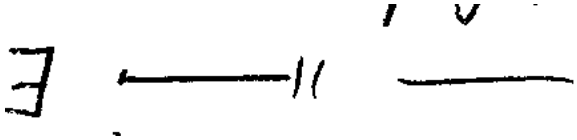
Let there be a Coxeter grp. and let $a = s_{i_1} \dots s_{i_\ell}$ be written minimally (as short as possible).

Uncertain reading. The thickened letter before $= s_{i_1} \dots s_{i_\ell}$ reads a.



Original source details: GELFAND-P001294

$a < b$ if $\exists$ a min. decomp. $a = w_{i_1} \dots w_{i_\ell}$.



Original source details: GELFAND-P001294

Diagram and typed description.

$\exists$ ——// —— $b = \dots.(w_{i_1} \dots w_{i_\ell}) \dots.$

(Bruhat order)

$a \overset{w}{<} b$ if $w^{-1}a < w^{-1}b$.



Uncertain reading. The thickened letter to the right of the $<$ marked w reads b.

Original source details: GELFAND-P001294

W is a Coxeter grp., $L \subset W$, Card $L < \infty$. L is called a flag matroid [wavy underlining] if for every w-ordering there is a min element in L.

One can enumerate all finite Coxeter grps. and the corresponding matroids.

GELFAND-P001295

A matroid of vectors is completely spec. by the list of max sets of lin. indep. vectors. [Both lines have wavy underlining.]

[Boxed and crossed out. Def. [crossed out] A parab. subgroup P of a Coxeter grp. is a subgroup gen. by a set $S \ldots$]



Uncertain reading. An illegible entry is cancelled after 'Def.'

Original source details: GELFAND-P001295

Is every matroid representable as a vector matroid?

Not every one!!

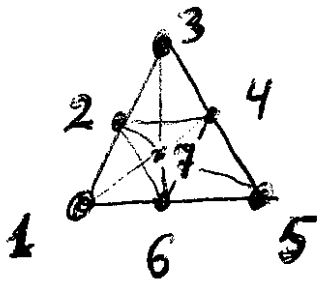
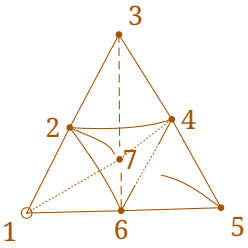


Diagram and typed description.



Some interior strokes break near point 7. Their exact joins are indistinct in places.

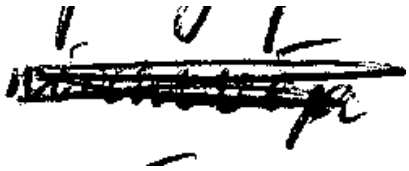
Original source details: GELFAND-P001295

$$\begin{array}{lll} 123 & 275 & 246 \\ 165 & 174 & \\ 345 & 376 & \end{array}$$

On the work of Beilinson and Ginzburg. 'Quadr. algebras'.

Consider a graded algebra with integer grading ($0, 1, 2, \ldots$ — degrees).

A_0 — [crossed out] a field;



Uncertain reading. Before 'field' a word is cancelled, possibly 'algebra'.

Original source details: GELFAND-P001295

Require that the algebra be generated by $A_1 = V$ ($\dim_{A_0} V < \infty$).

[crossed out] Relations?



Original source details: GELFAND-P001295

Uncertain reading. Before ‘Relations?’ is a cancelled entry, possibly ‘Зад.’.

[A broken vertical line at left.]

Example. $x_i x_k - x_k x_i = 0 \Rightarrow$ polynomial alg.

Requirement. All rel. are generated by elements of $V \otimes V$.

$$W \subset V \otimes V$$

(subspace.)

$$A_0 \oplus V \oplus (V \otimes V) \oplus \ldots \oplus V^{\otimes n} \oplus \ldots = T(V)$$



Original source details: GELFAND-P001295

Uncertain reading. A superscript, possibly n, is cancelled after V, before $\otimes n$.

Required. $\exists$ a lin. space $W \subset V \otimes V$ gen. all rel. in A.

$$(A = T(V)/J, \quad \text{where } J \text{ is gen. by } W$$
$$J = J(W))$$

Dual.

$$A_0 \oplus V^* \oplus V^* \otimes V^* \oplus \ldots \oplus (V^*)^{\otimes n} \oplus \ldots$$

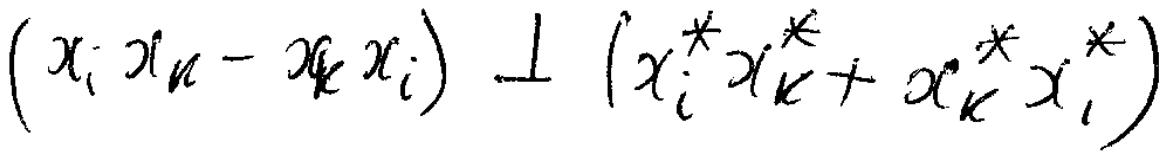
Consider in it $W^\perp \subset V^* \otimes V^*$ and the corresponding algebra

$$A^! = T(V^*)/J(W^\perp).$$

It is called the dual algebra. [Wavy underlining.]

Example 1. Let A be the commut. algebra of polynomials in $x_1, \ldots, x_n$. [‘polynomials’ has wavy underlining.]

$$(x_i x_n - x_k x_i) \perp (x_i^* x_k^* + x_k^* x_i^*)$$



Original source details: GELFAND-P001295

Uncertain reading. In the first product the second index reads n. In the next product the first index reads k.

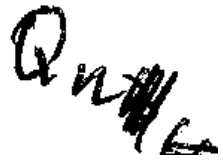
Rel. $\sum a_{ij}\, x_i \otimes x_j$ with $a_{ij} = -a_{ji}$.

$A^!$ is a Grassmann algebra.

GELFAND-P001296

[source number: 72]

Example 2. [Broken underlining.] Consider n quadr. forms $Q_1, \ldots, Q_n$ and the quotient of all commut. polynomials by $J\langle Q\rangle$, i.e.



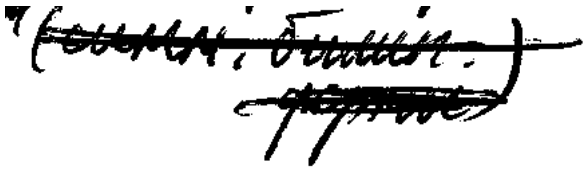
Original source details: GELFAND-P001296

Uncertain reading. An illegible mark is cancelled after Q_n.



Original source details: GELFAND-P001296

Uncertain reading. The delimiters around Q are overwritten. $J\langle Q\rangle$ follows the visible angle strokes. The formula below reads $J(Q)$.



Original source details: GELFAND-P001296

Uncertain reading. At right '(symm. bilin. forms)' is crossed out. The final word is heavily obscured.

Original source details: GELFAND-P001296

[Above the formula at right $T(V)$.]



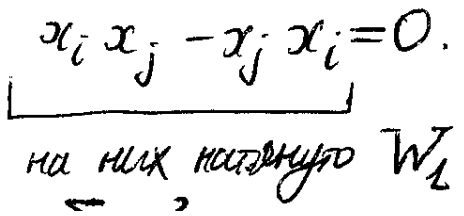
Original source details: GELFAND-P001296

Uncertain reading. The entry $T(V)$ stands by itself.

$$A = S(V)/J(Q)$$

For example. [Wavy underlining.]

$$\sum x_i^2 = 0,$$



Original source details: GELFAND-P001296

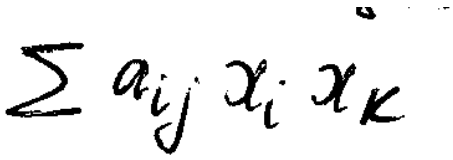
Diagram and typed description.

$x_i x_j - x_j x_i = 0.$

they span W_1

W is spanned by W_1 and $\sum x_i^2$.
Their lin. combinations?

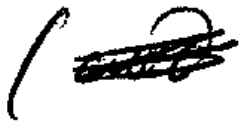
$$\sum a_{ij}x_ix_k, \qquad \begin{aligned} a_{ij} &= -a_{ji}, \quad i \neq j, \\ a_{ii} &= a_{jj}. \end{aligned}$$



Original source details: GELFAND-P001296

Uncertain reading. In the sum $\sum a_{ij}x_ix_k$ the second factor has index k.

$$W^{\perp} = \{b_{ij}x_i^*x_j^*; \quad b_{ij} = b_{ji}; \quad \sum b_{ii} = 0\}$$



Original source details: GELFAND-P001296

Uncertain reading. To the right of the formula is an opening parenthesis followed by an illegible cancelled entry.

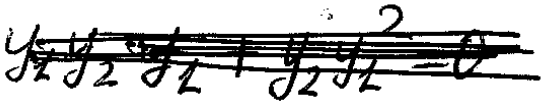
symm. matrices with trace 0.
Independent relations. [Wavy underlining.]

$$y_iy_j + y_jy_i = 0 \text{ for } i \neq j; \qquad y_i^2 = y_j^2 \quad (\forall i, j)$$

Put $y_i^2 = h$. Then

$$y_iy_j + y_jy_i = 2\delta_{ij}h.$$

[Crossed out $y_1y_2y_1 + y_2y_1^2 = 0$.]



Original source details: GELFAND-P001296

Uncertain reading. The crossed-out line reads $y_1y_2y_1 + y_2y_1^2 = 0$.

$$y_1^2y_i = y_iy_1^2 \qquad (\text{since } y_1^2 = y_i^2)$$

$\Rightarrow h \in$ the center! [Wavy underlining.]